PROBLEMS SHEET 3, PART 2 • OFFICIAL SOLUTIONS

Department of Mathematics • UC3M Aerospace Engineering

Chapter 3 Problem Set, Part 2: Forced Oscillations, Resonance and Variation of Constants

Complete, step-by-step solutions to the nine exercises of the second Chapter 3 sheet: eleven inhomogeneous initial value problems by undetermined coefficients, superposition, beats and resonance, the damped forced oscillator (amplitude and phase), higher-order equations, variation of constants, the structure of IVP solutions and reduction of order for inhomogeneous equations.

Source Document:	sources/cuatrimestre-1/04-advanced-maths/unit-03-second-order-linear-odes/problemas/probls_ch3-ii_2627.pdf

EXERCISE 1

Constant-Coefficient Equations with Forcing (Eleven IVPs)

Original statement · probls_ch3-ii_2627.pdf

Undetermined Coefficients • 11 IVPs • Resonance

STATEMENT

Find the general solution of the following differential equations. You will find it useful to look up Exercise 4 of Chapter 3, part 1. For each item (n), find the unique solution such that $x(0) = n$ and $\dot{x}(0) = 0$. E.g. for item (iii), $x(0) = 3$, etc.

- (i) $\ddot{x} - 4x = t^2$;
- (ii) $\ddot{x} - 4\dot{x} = t^2$;
- (iii) $\ddot{x} + \dot{x} - 2x = 3e^{-t}$;
- (iv) $\ddot{x} + \dot{x} - 2x = e^{t}$;
- (v) $\ddot{x} + 2\dot{x} + x = e^{-t}$;
- (vi) $\ddot{x} + \omega^2 x = \sin(\alpha t)$ for $\alpha \neq \omega$;
- (vii) $\ddot{x} + \omega^2 x = \sin(\alpha t)$ for $\alpha = \omega$;

- (viii) $\ddot{x} + 2\dot{x} + 10x = e^{-t}$;
- (ix) $\ddot{x} + 2\dot{x} + 10x = e^{-t}\cos(3t)$;
- (x) $\ddot{x} + 6\dot{x} + 10x = e^{-3t}\cos t$,
- (xi) $\ddot{x} + 4\dot{x} + 4x = e^{2t}$;

PHASE 1: SETUP, CLASSIFICATION AND DATA

All eleven equations are **linear, second-order, inhomogeneous equations with constant coefficients**, $L[x] = \ddot{x} + b\dot{x} + cx = f(t)$ (the coefficient of $\ddot{x}$ is 1). The forcing f is built from polynomials, exponentials and sinusoids, so the method of undetermined coefficients applies (Robinson Ch. 14). Hypotheses: for (vi) and (vii) the parameters $\omega > 0$ and α are real, with $\alpha \neq \omega$ in (vi) and $\alpha = \omega$ in (vii).

The coefficients are continuous on all of $\mathbb{R}$, so for every pair of initial data there is exactly one solution, defined for all t (existence and uniqueness, Robinson Thm 11.1 and Thm 16.1). The data of item (n) are $x(0) = n$, $\dot{x}(0) = 0$, with n the position of the item in the list:

Item	**Equation** $L[x] = f$	$x(0)$	$\dot{x}(0)$
(i)	$\ddot{x} - 4x = t^2$	1	0
(ii)	$\ddot{x} - 4\dot{x} = t^2$	2	0
(iii)	$\ddot{x} + \dot{x} - 2x = 3e^{-t}$	3	0
(iv)	$\ddot{x} + \dot{x} - 2x = e^{t}$	4	0
(v)	$\ddot{x} + 2\dot{x} + x = e^{-t}$	5	0
(vi)	$\ddot{x} + \omega^2 x = \sin(\alpha t)$ for $\alpha \neq \omega$	6	0
(vii)	$\ddot{x} + \omega^2 x = \sin(\alpha t)$ for $\alpha = \omega$	7	0
(viii)	$\ddot{x} + 2\dot{x} + 10x = e^{-t}$	8	0
(ix)	$\ddot{x} + 2\dot{x} + 10x = e^{-t}\cos(3t)$	9	0
(x)	$\ddot{x} + 6\dot{x} + 10x = e^{-3t}\cos t$	10	0
(xi)	$\ddot{x} + 4\dot{x} + 4x = e^{2t}$	11	0

Strategy (Robinson Sec. 14.1, Lemma): the general solution of $L[x] = f$ is $x = x_h + x_p$, with x_h the general solution of $L[x] = 0$ (two arbitrary constants c_1, c_2) and x_p any one particular solution. The constants are then fixed by the two initial conditions.

PHASE 2: METHOD, NOTATION AND THE GUESS FOR EACH ITEM

Write $P(k) = k^2 + bk + c$ for the characteristic polynomial. A forcing $c_0\, t^m e^{\sigma t}\{\cos \omega t, \sin \omega t\}$ is, for the purpose of the guess, the exponent $s = \sigma + i\omega$ (and $s = \sigma$ when there is no sinusoid; $s = 0$ for a polynomial). Rule of thumb (Robinson Sec. 14.5, p. 139): take the standard guess of the same form as f with undetermined coefficients, and multiply it by t^{m_s} where m_s is the multiplicity of s as a root of P ($m_s = 0$ when $P(s) \neq 0$).

Forcing	**Standard guess x_p**	**If s is a root of P of multiplicity m_s**
polynomial of degree m	$K_0 + K_1 t + \cdots + K_m t^m$	multiply by t^{m_s}
$e^{\sigma t}$	$K e^{\sigma t}$	$K t^{m_s} e^{\sigma t}$
$\sin \omega t$ or $\cos \omega t$	$K_1 \cos \omega t + K_2 \sin \omega t$ (both, always)	$t^{m_s}(K_1 \cos \omega t + K_2 \sin \omega t)$
$e^{\sigma t} \cos \omega t$	$e^{\sigma t}(K_1 \cos \omega t + K_2 \sin \omega t)$	$t^{m_s} e^{\sigma t}(K_1 \cos \omega t + K_2 \sin \omega t)$

Applying the rule item by item (the roots of P were found in Exercise 4 of part 1 for the homogeneous equations; the last column is m_s):

Item	$P(k)$	**Roots of P**	**Exponent s of f**	m_s
(i)	$k^2 - 4$	$-2, 2$	0	0
(ii)	$k^2 - 4k$	$0, 4$	0	1
(iii)	$k^2 + k - 2$	$-2, 1$	-1	0
(iv)	$k^2 + k - 2$	$-2, 1$	1	1
(v)	$k^2 + 2k + 1$	-1 (double)	-1	2
(vi)	$k^2 + \omega^2$	$-i\omega, i\omega$	$i\alpha$	0
(vii)	$k^2 + \omega^2$	$-i\omega, i\omega$	$i\omega$	1
(viii)	$k^2 + 2k + 10$	$-1 - 3i, -1 + 3i$	-1	0
(ix)	$k^2 + 2k + 10$	$-1 - 3i, -1 + 3i$	$-1 + 3i$	1
(x)	$k^2 + 6k + 10$	$-3 - i, -3 + i$	$-3 + i$	1
(xi)	$k^2 + 4k + 4$	-2 (double)	2	0

Items with $m_s \geq 1$ (resonant forcing, marked by a non-zero last column) need the extra factor t^{m_s}: (ii), (iv), (v), (vii), (ix), (x).

`PHASE 3: STEP-BY-STEP DERIVATION OF THE ELEVEN SOLUTIONS`

(i) $\ddot{x} - 4x = t^2$, with $x(0) = 1$, $\dot{x}(0) = 0$

The homogeneous equation $\ddot{x} - 4x = 0$ has constant coefficients, so we try $x = e^{kt}$ (Robinson Sec. 12.1 and 16.1): by the chain rule each derivative multiplies by k, and dividing by $e^{kt} \neq 0$ leaves the characteristic equation

$$P(k) = k^2 - 4 = 0$$

Its discriminant is $\Delta = b^2 - 4ac = (0)^2 - 4(1)(-4) = 16$.

Factorising, $P(k) = (k - 2)\,(k + 2)$, so the roots are

$$k = -2, \qquad k = 2$$

A real root k of multiplicity m gives $t^j e^{kt}$ for $j = 0, \dots, m-1$; a complex pair $\rho \pm i\omega$ gives $t^j e^{\rho t}\cos \omega t$ and $t^j e^{\rho t}\sin \omega t$ (Euler's formula, Robinson Sec. 12.3). The fundamental set is e^{-2t}, e^{2t}, so the complementary function is

$$x_h(t) = c_1 e^{-2t} + c_2 e^{2t}$$

The forcing t^2 is a polynomial of degree 2; a polynomial is e^{0t} times a polynomial, so the exponent to test is $s = 0$: $P(0) = -4$.

Since $P(s) \neq 0$, no term of the standard guess solves the homogeneous equation, so the standard guess is used as it stands.

Guess for the particular integral, with undetermined coefficients:

$$x_p(t) = K_1 + K_2 t + K_3 t^2$$

Differentiate term by term (product rule on $t^j e^{\sigma t}$, chain rule on $\cos \omega t$ and $\sin \omega t$):

$$\begin{aligned} x_p'(t) &= K_2 + 2K_3 t \\ x_p''(t) &= 2K_3 \end{aligned}$$

Substitute into $L[x_p] = x_p'' - 4x_p$ and collect the terms that multiply the same function:

$$L[x_p] = -4K_3\, t^2 - 4K_2\, t + (-4K_1 + 2K_3)$$

This must equal the forcing $f(t) = t^2$ for every t. The functions on the left are linearly independent, so the coefficient of each must match the coefficient of the same function in f:

$$\begin{aligned} t^2: &\quad -4K_3 = 1 \\ t: &\quad -4K_2 = 0 \\ 1: &\quad -4K_1 + 2K_3 = 0 \end{aligned}$$

The linear system is

$$
\begin{aligned}
-4K_3 &= 1 \quad (1)\\
-4K_2 &= 0 \quad (2)\\
-4K_1 + 2K_3 &= 0 \quad (3)
\end{aligned}
$$

$-4K_3 = 1$ contains one unknown only, so $K_3 = -\frac{1}{4}$.

Substitute in $-4K_1 + 2K_3 = 0$: $-4K_1 - \frac{1}{2} = 0$.

$-4K_2 = 0$ contains one unknown only, so $K_2 = 0$.

$-4K_1 - \frac{1}{2} = 0$ contains one unknown only, so $K_1 = -\frac{1}{8}$.

Hence $K_1 = -\frac{1}{8}, K_2 = 0, K_3 = -\frac{1}{4}$.

Therefore

$$
x_p(t) = -\frac{t^2}{4} - \frac{1}{8}
$$

The general solution is the complementary function plus the particular integral:

$$
x(t) = x_h(t) + x_p(t) = c_1 e^{-2t} + c_2 e^{2t} - \frac{t^2}{4} - \frac{1}{8}
$$

Initial conditions. Differentiate the general solution (product and chain rules):

$$
x'(t) = -2c_1 e^{-2t} + 2c_2 e^{2t} - \frac{t}{2}
$$

Evaluate at $t = 0$, using $e^0 = 1$, $\cos 0 = 1$ and $\sin 0 = 0$:

$$
\begin{aligned}
x(0) &= c_1 + c_2 - \frac{1}{8} = 1\\
x'(0) &= -2c_1 + 2c_2 = 0
\end{aligned}
$$

The linear system is

$$
\begin{aligned}
c_1 + c_2 - \frac{1}{8} &= 1 \quad (1)\\
-2c_1 + 2c_2 &= 0 \quad (2)
\end{aligned}
$$

Solve $c_1 + c_2 - \frac{1}{8} = 1$ for c_1: $c_1 = \frac{9}{8} - c_2$.

Substitute in $-2c_1 + 2c_2 = 0$: $4c_2 - \frac{9}{4} = 0$.

$4c_2 - \frac{9}{4} = 0$ contains one unknown only, so $c_2 = \frac{9}{16}$.

Back-substitution: $c_1 = \frac{9}{8} - c_2 = \frac{9}{16}$.

Hence $c_1 = \frac{9}{16}, c_2 = \frac{9}{16}$.

Substituting the constants into the general solution,

$$\boxed{x(t) = -\frac{t^2}{4} + \frac{9e^{2t}}{16} - \frac{1}{8} + \frac{9e^{-2t}}{16}}$$

(ii) $\ddot{x} - 4\dot{x} = t^2$**, with** $x(0) = 2,\ \dot{x}(0) = 0$

The homogeneous equation $\ddot{x} - 4\dot{x} = 0$ has constant coefficients, so we try $x = e^{kt}$ (Robinson Sec. 12.1 and 16.1): by the chain rule each derivative multiplies by k, and dividing by $e^{kt} \neq 0$ leaves the characteristic equation

$$P(k) = k^2 - 4k = 0$$

Its discriminant is $\Delta = b^2 - 4ac = (-4)^2 - 4(1)(0) = 16$.

Factorising, $P(k) = k\,(k - 4)$, so the roots are

$$k = 0, \qquad k = 4$$

A real root k of multiplicity m gives $t^j e^{kt}$ for $j = 0, \ldots, m-1$; a complex pair $\rho \pm i\omega$ gives $t^j e^{\rho t}\cos \omega t$ and $t^j e^{\rho t}\sin \omega t$ (Euler's formula, Robinson Sec. 12.3). The fundamental set is $1, e^{4t}$, so the complementary function is

$$x_h(t) = c_1 + c_2 e^{4t}$$

The forcing t^2 is a polynomial of degree 2; a polynomial is e^{0t} times a polynomial, so the exponent to test is $s = 0$: $P(0) = 0$.

Since $P(s) = 0$ and s is a root of multiplicity 1, the standard guess would contain terms of the complementary function. By the rule of thumb (Robinson Sec. 14.5, p. 139) the guess is multiplied by t^1.

Guess for the particular integral, with undetermined coefficients:

$$x_p(t) = K_1 t + K_2 t^2 + K_3 t^3$$

Differentiate term by term (product rule on $t^j e^{\sigma t}$, chain rule on $\cos \omega t$ and $\sin \omega t$):

$$\begin{aligned} x_p'(t) &= K_1 + 2K_2 t + 3K_3 t^2 \\ x_p''(t) &= 2K_2 + 6K_3 t \end{aligned}$$

Substitute into $L[x_p] = x_p'' - 4x_p'$ and collect the terms that multiply the same function:

$$L[x_p] = -12K_3\, t^2 + (-8K_2 + 6K_3)\, t + (-4K_1 + 2K_2)$$

This must equal the forcing $f(t) = t^2$ for every t. The functions on the left are linearly independent, so the coefficient of each must match the coefficient of the same function in f:

$$\begin{aligned} t^2: &\quad -12K_3 = 1 \\ t: &\quad -8K_2 + 6K_3 = 0 \\ 1: &\quad -4K_1 + 2K_2 = 0 \end{aligned}$$

The linear system is

$$\begin{aligned} -12K_3 &= 1 \quad (1) \\ -8K_2 + 6K_3 &= 0 \quad (2) \\ -4K_1 + 2K_2 &= 0 \quad (3) \end{aligned}$$

$-12K_3 = 1$ contains one unknown only, so $K_3 = -\frac{1}{12}$.

Substitute in $-8K_2 + 6K_3 = 0$: $-8K_2 - \frac{1}{2} = 0$.

$-8K_2 - \frac{1}{2} = 0$ contains one unknown only, so $K_2 = -\frac{1}{16}$.

Substitute in $-4K_1 + 2K_2 = 0$: $-4K_1 - \frac{1}{8} = 0$.

$-4K_1 - \frac{1}{8} = 0$ contains one unknown only, so $K_1 = -\frac{1}{32}$.

Hence $K_1 = -\frac{1}{32}, K_2 = -\frac{1}{16}, K_3 = -\frac{1}{12}$.

Therefore

$$x_p(t) = -\frac{t^3}{12} - \frac{t^2}{16} - \frac{t}{32}$$

The general solution is the complementary function plus the particular integral:

$$x(t) = x_h(t) + x_p(t) = c_1 + c_2 e^{4t} - \frac{t^3}{12} - \frac{t^2}{16} - \frac{t}{32}$$

Initial conditions. Differentiate the general solution (product and chain rules):

$$x'(t) = 4c_2 e^{4t} - \frac{t^2}{4} - \frac{t}{8} - \frac{1}{32}$$

Evaluate at $t = 0$, using $e^0 = 1$, $\cos 0 = 1$ and $\sin 0 = 0$:

$$x(0) = c_1 + c_2 = 2$$
$$x'(0) = 4c_2 - \frac{1}{32} = 0$$

The linear system is

$$c_1 + c_2 = 2 \quad (1)$$
$$4c_2 - \frac{1}{32} = 0 \quad (2)$$

$4c_2 - \frac{1}{32} = 0$ contains one unknown only, so $c_2 = \frac{1}{128}$.

Substitute in $c_1 + c_2 = 2$: $c_1 + \frac{1}{128} = 2$.

$c_1 + \frac{1}{128} = 2$ contains one unknown only, so $c_1 = \frac{255}{128}$.

Hence $c_1 = \frac{255}{128}, c_2 = \frac{1}{128}$.

Substituting the constants into the general solution,

$$\boxed{x(t) = -\frac{t^3}{12} - \frac{t^2}{16} - \frac{t}{32} + \frac{e^{4t}}{128} + \frac{255}{128}}$$

(iii) $\ddot{x} + \dot{x} - 2x = 3e^{-t}$**, with** $x(0) = 3,\ \dot{x}(0) = 0$

The homogeneous equation $\ddot{x} + \dot{x} - 2x = 0$ has constant coefficients, so we try $x = e^{kt}$ (Robinson Sec. 12.1 and 16.1): by the chain rule each derivative multiplies by k, and dividing by $e^{kt} \neq 0$ leaves the characteristic equation

$$P(k) = k^2 + k - 2 = 0$$

Its discriminant is $\Delta = b^2 - 4ac = (1)^2 - 4(1)(-2) = 9$.

Factorising, $P(k) = (k - 1)\,(k + 2)$, so the roots are

$$k = -2, \qquad k = 1$$

A real root k of multiplicity m gives $t^j e^{kt}$ for $j = 0, \dots, m-1$; a complex pair $\rho \pm i\omega$ gives $t^j e^{\rho t}\cos\omega t$ and $t^j e^{\rho t}\sin\omega t$ (Euler's formula, Robinson Sec. 12.3). The fundamental set is e^{-2t}, e^{t}, so the complementary function is

$$x_h(t) = c_1 e^{-2t} + c_2 e^{t}$$

The forcing $3e^{-t}$ is an exponential with exponent $s = -1$; we test it in the characteristic polynomial: $P(-1) = -2$.

Since $P(s) \neq 0$, no term of the standard guess solves the homogeneous equation, so the standard guess is used as it stands.

Guess for the particular integral, with undetermined coefficients:

$$x_p(t) = K_1 e^{-t}$$

Differentiate term by term (product rule on $t^j e^{\sigma t}$, chain rule on $\cos\omega t$ and $\sin\omega t$):

$$\begin{aligned} x_p'(t) &= -K_1 e^{-t} \\ x_p''(t) &= K_1 e^{-t} \end{aligned}$$

Substitute into $L[x_p] = x_p'' + x_p' - 2x_p$ and collect the terms that multiply the same function:

$$L[x_p] = -2K_1\, e^{-t}$$

This must equal the forcing $f(t) = 3e^{-t}$ for every t. The functions on the left are linearly independent, so the coefficient of each must match the coefficient of the same function in f:

$$e^{-t}: \quad -2K_1 = 3$$

The linear system is

$$-2K_1 = 3 \quad (1)$$

$-2K_1 = 3$ contains one unknown only, so $K_1 = -\frac{3}{2}$.

Hence $K_1 = -\frac{3}{2}$.

Therefore

$$x_p(t) = -\frac{3e^{-t}}{2}$$

The general solution is the complementary function plus the particular integral:

$$x(t) = x_h(t) + x_p(t) = c_1 e^{-2t} + c_2 e^{t} - \frac{3e^{-t}}{2}$$

Initial conditions. Differentiate the general solution (product and chain rules):

$$x'(t) = -2c_1 e^{-2t} + c_2 e^{t} + \frac{3e^{-t}}{2}$$

Evaluate at $t = 0$, using $e^0 = 1$, $\cos 0 = 1$ and $\sin 0 = 0$:

$$x(0) = c_1 + c_2 - \frac{3}{2} = 3$$
$$x'(0) = -2c_1 + c_2 + \frac{3}{2} = 0$$

The linear system is

$$c_1 + c_2 - \frac{3}{2} = 3 \quad (1)$$
$$-2c_1 + c_2 + \frac{3}{2} = 0 \quad (2)$$

Solve $c_1 + c_2 - \frac{3}{2} = 3$ for c_1: $c_1 = \frac{9}{2} - c_2$.

Substitute in $-2c_1 + c_2 + \frac{3}{2} = 0$: $3c_2 - \frac{15}{2} = 0$.

$3c_2 - \frac{15}{2} = 0$ contains one unknown only, so $c_2 = \frac{5}{2}$.

Back-substitution: $c_1 = \frac{9}{2} - c_2 = 2$.

Hence $c_1 = 2$, $c_2 = \frac{5}{2}$.

Substituting the constants into the general solution,

$$\boxed{x(t) = \frac{5e^{t}}{2} - \frac{3e^{-t}}{2} + 2e^{-2t}}$$

(iv) $\ddot{x} + \dot{x} - 2x = e^{t}$**, with** $x(0) = 4,\ \dot{x}(0) = 0$

The homogeneous equation $\ddot{x} + \dot{x} - 2x = 0$ has constant coefficients, so we try $x = e^{kt}$ (Robinson Sec. 12.1 and 16.1): by the chain rule each derivative multiplies by k, and dividing by $e^{kt} \neq 0$ leaves the characteristic equation

$$P(k) = k^2 + k - 2 = 0$$

Its discriminant is $\Delta = b^2 - 4ac = (1)^2 - 4(1)(-2) = 9$.

Factorising, $P(k) = (k - 1)\,(k + 2)$, so the roots are

$$k = -2, \qquad k = 1$$

A real root k of multiplicity m gives $t^{j}e^{kt}$ for $j = 0, \ldots, m-1$; a complex pair $\rho \pm i\omega$ gives $t^{j}e^{\rho t}\cos\omega t$ and $t^{j}e^{\rho t}\sin\omega t$ (Euler's formula, Robinson Sec. 12.3). The fundamental set is e^{-2t}, e^{t}, so the complementary function is

$$x_h(t) = c_1 e^{-2t} + c_2 e^{t}$$

The forcing e^{t} is an exponential with exponent $s = 1$; we test it in the characteristic polynomial: $P(1) = 0$.

Since $P(s) = 0$ and s is a root of multiplicity 1, the standard guess would contain terms of the complementary function. By the rule of thumb (Robinson Sec. 14.5, p. 139) the guess is multiplied by t^{1}.

Guess for the particular integral, with undetermined coefficients:

$$x_p(t) = K_1 t e^t$$

Differentiate term by term (product rule on $t^j e^{\sigma t}$, chain rule on $\cos \omega t$ and $\sin \omega t$):

$$x_p'(t) = K_1 t e^t + K_1 e^t$$
$$x_p''(t) = K_1 t e^t + 2K_1 e^t$$

Substitute into $L[x_p] = x_p'' + x_p' - 2x_p$ and collect the terms that multiply the same function:

$$L[x_p] = 3K_1\, e^t$$

This must equal the forcing $f(t) = e^t$ for every t. The functions on the left are linearly independent, so the coefficient of each must match the coefficient of the same function in f:

$$e^t: \quad 3K_1 = 1$$

The linear system is

$$3K_1 = 1 \quad (1)$$

$3K_1 = 1$ contains one unknown only, so $K_1 = \frac{1}{3}$.

Hence $K_1 = \frac{1}{3}$.

Therefore

$$x_p(t) = \frac{te^t}{3}$$

The general solution is the complementary function plus the particular integral:

$$x(t) = x_h(t) + x_p(t) = c_1 e^{-2t} + c_2 e^t + \frac{te^t}{3}$$

Initial conditions. Differentiate the general solution (product and chain rules):

$$x'(t) = -2c_1 e^{-2t} + c_2 e^{t} + \frac{te^{t}}{3} + \frac{e^{t}}{3}$$

Evaluate at $t = 0$, using $e^{0} = 1$, $\cos 0 = 1$ and $\sin 0 = 0$:

$$x(0) = c_1 + c_2 = 4$$
$$x'(0) = -2c_1 + c_2 + \frac{1}{3} = 0$$

The linear system is

$$c_1 + c_2 = 4 \quad (1)$$
$$-2c_1 + c_2 + \frac{1}{3} = 0 \quad (2)$$

Solve $c_1 + c_2 = 4$ for c_1: $c_1 = 4 - c_2$.

Substitute in $-2c_1 + c_2 + \frac{1}{3} = 0$: $3c_2 - \frac{23}{3} = 0$.

$3c_2 - \frac{23}{3} = 0$ contains one unknown only, so $c_2 = \frac{23}{9}$.

Back-substitution: $c_1 = 4 - c_2 = \frac{13}{9}$.

Hence $c_1 = \frac{13}{9}$, $c_2 = \frac{23}{9}$.

Substituting the constants into the general solution,

$$\boxed{x(t) = \frac{te^{t}}{3} + \frac{23e^{t}}{9} + \frac{13e^{-2t}}{9}}$$

(v) $\ddot{x} + 2\dot{x} + x = e^{-t}$**, with** $x(0) = 5,\ \dot{x}(0) = 0$

The homogeneous equation $\ddot{x} + 2\dot{x} + x = 0$ has constant coefficients, so we try $x = e^{kt}$ (Robinson Sec. 12.1 and 16.1): by the chain rule each derivative multiplies by k, and dividing by $e^{kt} \neq 0$ leaves the characteristic equation

$$P(k) = k^2 + 2k + 1 = 0$$

Its discriminant is $\Delta = b^2 - 4ac = (2)^2 - 4(1)(1) = 0$.

Factorising, $P(k) = (k+1)^2$, so the roots are

$$k = -1 \text{ (multiplicity } 2)$$

A real root k of multiplicity m gives $t^j e^{kt}$ for $j = 0, \dots, m-1$; a complex pair $\rho \pm i\omega$ gives $t^j e^{\rho t}\cos \omega t$ and $t^j e^{\rho t}\sin \omega t$ (Euler's formula, Robinson Sec. 12.3). The fundamental set is e^{-t}, te^{-t}, so the complementary function is

$$x_h(t) = c_1 e^{-t} + c_2 t e^{-t}$$

The forcing e^{-t} is an exponential with exponent $s = -1$; we test it in the characteristic polynomial: $P(-1) = 0$.

Since $P(s) = 0$ and s is a root of multiplicity 2, the standard guess would contain terms of the complementary function. By the rule of thumb (Robinson Sec. 14.5, p. 139) the guess is multiplied by t^2.

Guess for the particular integral, with undetermined coefficients:

$$x_p(t) = K_1 t^2 e^{-t}$$

Differentiate term by term (product rule on $t^j e^{\sigma t}$, chain rule on $\cos \omega t$ and $\sin \omega t$):

$$\begin{aligned} x_p'(t) &= -K_1 t^2 e^{-t} + 2K_1 t e^{-t} \\ x_p''(t) &= K_1 t^2 e^{-t} - 4K_1 t e^{-t} + 2K_1 e^{-t} \end{aligned}$$

Substitute into $L[x_p] = x_p'' + 2x_p' + x_p$ and collect the terms that multiply the same function:

$$L[x_p] = 2K_1\, e^{-t}$$

This must equal the forcing $f(t) = e^{-t}$ for every t. The functions on the left are linearly independent, so the coefficient of each must match the coefficient of the same function in f:

$$e^{-t}: \quad 2K_1 = 1$$

The linear system is

$$2K_1 = 1 \quad (1)$$

$2K_1 = 1$ contains one unknown only, so $K_1 = \frac{1}{2}$.

Hence $K_1 = \frac{1}{2}$.

Therefore

$$x_p(t) = \frac{t^2 e^{-t}}{2}$$

The general solution is the complementary function plus the particular integral:

$$x(t) = x_h(t) + x_p(t) = c_1 e^{-t} + c_2 t e^{-t} + \frac{t^2 e^{-t}}{2}$$

Initial conditions. Differentiate the general solution (product and chain rules):

$$x'(t) = -c_1 e^{-t} - c_2 t e^{-t} + c_2 e^{-t} - \frac{t^2 e^{-t}}{2} + t e^{-t}$$

Evaluate at $t = 0$, using $e^0 = 1$, $\cos 0 = 1$ and $\sin 0 = 0$:

$$x(0) = c_1 = 5$$
$$x'(0) = -c_1 + c_2 = 0$$

The linear system is

$$c_1 = 5 \quad (1)$$
$$-c_1 + c_2 = 0 \quad (2)$$

$c_1 = 5$ contains one unknown only, so $c_1 = 5$.

Substitute in $-c_1 + c_2 = 0$: $c_2 - 5 = 0$.

$c_2 - 5 = 0$ contains one unknown only, so $c_2 = 5$.

Hence $c_1 = 5, c_2 = 5$.

Substituting the constants into the general solution,

$$\boxed{x(t) = \frac{t^2 e^{-t}}{2} + 5te^{-t} + 5e^{-t}}$$

(vi) $\ddot{x} + \omega^2 x = \sin \alpha t$**, with** $x(0) = 6,\ \dot{x}(0) = 0$

The homogeneous equation $\ddot{x} + \omega^2 x = 0$ has constant coefficients, so we try $x = e^{kt}$ (Robinson Sec. 12.1 and 16.1): by the chain rule each derivative multiplies by k, and dividing by $e^{kt} \neq 0$ leaves the characteristic equation

$$P(k) = k^2 + \omega^2 = 0$$

Its discriminant is $\Delta = b^2 - 4ac = (0)^2 - 4(1)(\omega^2) = -4\omega^2$.

Factorising, $P(k) = (k - i\omega)\,(k + i\omega)$, so the roots are

$$k = -i\omega, \qquad k = i\omega$$

A real root k of multiplicity m gives $t^j e^{kt}$ for $j = 0, \ldots, m-1$; a complex pair $\rho \pm i\omega$ gives $t^j e^{\rho t} \cos \omega t$ and $t^j e^{\rho t} \sin \omega t$ (Euler's formula, Robinson Sec. 12.3). The fundamental set is $\cos \omega t$, $\sin \omega t$, so the complementary function is

$$x_h(t) = c_1 \cos \omega t + c_2 \sin \omega t$$

The forcing $\sin \alpha t$ is a sinusoidal term: it is the real or imaginary part of $e^{(i\alpha)t}$, so we test $s = i\alpha$ in the characteristic polynomial: $P(i\alpha) = -\alpha^2 + \omega^2$.

Since $P(s) \neq 0$, no term of the standard guess solves the homogeneous equation, so the standard guess is used as it stands.

Guess for the particular integral, with undetermined coefficients:

$$x_p(t) = K_1 \cos \alpha t + K_2 \sin \alpha t$$

Differentiate term by term (product rule on $t^j e^{\sigma t}$, chain rule on $\cos \omega t$ and $\sin \omega t$):

$$x_p'(t) = -K_1 \alpha \sin \alpha t + K_2 \alpha \cos \alpha t$$
$$x_p''(t) = -K_1 \alpha^2 \cos \alpha t - K_2 \alpha^2 \sin \alpha t$$

Substitute into $L[x_p] = x_p'' + \omega^2 x_p$ and collect the terms that multiply the same function:

$$L[x_p] = \left(-K_1\alpha^2 + K_1\omega^2\right)\,\cos\alpha\,t + \left(-K_2\alpha^2 + K_2\omega^2\right)\,\sin\alpha\,t$$

This must equal the forcing $f(t) = \sin\alpha t$ for every t. The functions on the left are linearly independent, so the coefficient of each must match the coefficient of the same function in f:

$$\begin{aligned}\cos\alpha\,t: &\quad -K_1\alpha^2 + K_1\omega^2 = 0\\ \sin\alpha\,t: &\quad -K_2\alpha^2 + K_2\omega^2 = 1\end{aligned}$$

The linear system is

$$\begin{aligned}-K_1\alpha^2 + K_1\omega^2 &= 0 \quad (1)\\ -K_2\alpha^2 + K_2\omega^2 &= 1 \quad (2)\end{aligned}$$

$-K_1\alpha^2 + K_1\omega^2 = 0$ contains one unknown only, so $K_1 = 0$.

$-K_2\alpha^2 + K_2\omega^2 = 1$ contains one unknown only, so $K_2 = -\frac{1}{\alpha^2-\omega^2}$.

Hence $K_1 = 0$, $K_2 = -\frac{1}{\alpha^2-\omega^2}$.

Therefore

$$x_p(t) = -\frac{\sin\alpha t}{\alpha^2 - \omega^2}$$

The general solution is the complementary function plus the particular integral:

$$x(t) = x_h(t) + x_p(t) = c_1\cos\omega t + c_2\sin\omega t - \frac{\sin\alpha t}{\alpha^2 - \omega^2}$$

Initial conditions. Differentiate the general solution (product and chain rules):

$$x'(t) = -\frac{\alpha\cos\alpha t}{\alpha^2 - \omega^2} - c_1\omega\sin\omega t + c_2\omega\cos\omega t$$

Evaluate at $t = 0$, using $e^0 = 1$, $\cos 0 = 1$ and $\sin 0 = 0$:

$$x(0) = c_1 = 6$$
$$x'(0) = -\frac{\alpha}{\alpha^2 - \omega^2} + c_2\omega = 0$$

The linear system is

$$c_1 = 6 \quad (1)$$
$$-\frac{\alpha}{\alpha^2 - \omega^2} + c_2\omega = 0 \quad (2)$$

$c_1 = 6$ contains one unknown only, so $c_1 = 6$.

$-\frac{\alpha}{\alpha^2-\omega^2} + c_2\omega = 0$ contains one unknown only, so $c_2 = \frac{\alpha}{\alpha^2\omega-\omega^3}$.

Hence $c_1 = 6$, $c_2 = \frac{\alpha}{\alpha^2\omega-\omega^3}$.

Substituting the constants into the general solution,

$$\boxed{x(t) = \frac{\alpha \sin \omega t}{\alpha^2\omega - \omega^3} + 6\cos \omega t - \frac{\sin \alpha t}{\alpha^2 - \omega^2}}$$

Writing the same solution with the denominator $\omega^2 - \alpha^2$ that appears naturally in the guess (check: $-\frac{1}{\alpha^2-\omega^2} = \frac{1}{\omega^2-\alpha^2}$ and $\frac{\alpha}{\alpha^2\omega-\omega^3} = -\frac{\alpha}{\omega(\omega^2-\alpha^2)}$):

$$\boxed{x(t) = 6\cos \omega t + \frac{\sin \alpha t - \dfrac{\alpha}{\omega}\sin \omega t}{\omega^2 - \alpha^2}}$$

The first term carries the initial displacement $x(0) = 6$; the fraction vanishes at $t = 0$ and has zero slope there ($\alpha\cos 0 - \alpha\cos 0 = 0$ after dividing by $\omega^2 - \alpha^2$), so the initial conditions hold.

(vii) $\ddot{x} + \omega^2 x = \sin \omega t$**, with** $x(0) = 7,\ \dot{x}(0) = 0$

The homogeneous equation $\ddot{x} + \omega^2 x = 0$ has constant coefficients, so we try $x = e^{kt}$ (Robinson Sec. 12.1 and 16.1): by the chain rule each derivative multiplies by k, and dividing by $e^{kt} \neq 0$ leaves the characteristic equation

$$P(k) = k^2 + \omega^2 = 0$$

Its discriminant is $\Delta = b^2 - 4ac = (0)^2 - 4(1)(\omega^2) = -4\omega^2$.

Factorising, $P(k) = (k - i\omega)\,(k + i\omega)$, so the roots are

$$k = -i\omega, \qquad k = i\omega$$

A real root k of multiplicity m gives $t^j e^{kt}$ for $j = 0, \dots, m-1$; a complex pair $\rho \pm i\omega$ gives $t^j e^{\rho t}\cos\omega t$ and $t^j e^{\rho t}\sin\omega t$ (Euler's formula, Robinson Sec. 12.3). The fundamental set is $\cos\omega t$, $\sin\omega t$, so the complementary function is

$$x_h(t) = c_1\cos\omega t + c_2\sin\omega t$$

The forcing $\sin\omega t$ is a sinusoidal term: it is the real or imaginary part of $e^{(i\omega)t}$, so we test $s = i\omega$ in the characteristic polynomial: $P(i\omega) = 0$.

Since $P(s) = 0$ and s is a root of multiplicity 1, the standard guess would contain terms of the complementary function. By the rule of thumb (Robinson Sec. 14.5, p. 139) the guess is multiplied by t^1.

Guess for the particular integral, with undetermined coefficients:

$$x_p(t) = K_1 t\cos\omega t + K_2 t\sin\omega t$$

Differentiate term by term (product rule on $t^j e^{\sigma t}$, chain rule on $\cos\omega t$ and $\sin\omega t$):

$$\begin{aligned} x_p'(t) &= -K_1\omega t\sin\omega t + K_1\cos\omega t + K_2\omega t\cos\omega t + K_2\sin\omega t \\ x_p''(t) &= -K_1\omega^2 t\cos\omega t - 2K_1\omega\sin\omega t - K_2\omega^2 t\sin\omega t + 2K_2\omega\cos\omega t \end{aligned}$$

Substitute into $L[x_p] = x_p'' + \omega^2 x_p$ and collect the terms that multiply the same function:

$$L[x_p] = 2K_2\omega\,\cos\omega\,t - 2K_1\omega\,\sin\omega\,t$$

This must equal the forcing $f(t) = \sin\omega t$ for every t. The functions on the left are linearly independent, so the coefficient of each must match the coefficient of the same function in f:

$$\begin{aligned} \cos\omega\,t: &\quad 2K_2\omega = 0 \\ \sin\omega\,t: &\quad -2K_1\omega = 1 \end{aligned}$$

The linear system is

$$\begin{aligned} 2K_2\omega &= 0 \quad (1) \\ -2K_1\omega &= 1 \quad (2) \end{aligned}$$

$2K_2\omega = 0$ contains one unknown only, so $K_2 = 0$.

$-2K_1\omega = 1$ contains one unknown only, so $K_1 = -\frac{1}{2\omega}$.

Hence $K_1 = -\frac{1}{2\omega}$, $K_2 = 0$.

Therefore

$$x_p(t) = -\frac{t\cos\omega t}{2\omega}$$

The general solution is the complementary function plus the particular integral:

$$x(t) = x_h(t) + x_p(t) = c_1\cos\omega t + c_2\sin\omega t - \frac{t\cos\omega t}{2\omega}$$

Initial conditions. Differentiate the general solution (product and chain rules):

$$x'(t) = -c_1\omega\sin\omega t + c_2\omega\cos\omega t + \frac{t\sin\omega t}{2} - \frac{\cos\omega t}{2\omega}$$

Evaluate at $t = 0$, using $e^0 = 1$, $\cos 0 = 1$ and $\sin 0 = 0$:

$$\begin{aligned} x(0) &= c_1 = 7 \\ x'(0) &= c_2\omega - \frac{1}{2\omega} = 0 \end{aligned}$$

The linear system is

$$\begin{aligned} c_1 &= 7 \quad (1) \\ c_2\omega - \frac{1}{2\omega} &= 0 \quad (2) \end{aligned}$$

$c_1 = 7$ contains one unknown only, so $c_1 = 7$.

$c_2\omega - \frac{1}{2\omega} = 0$ contains one unknown only, so $c_2 = \frac{1}{2\omega^2}$.

Hence $c_1 = 7$, $c_2 = \frac{1}{2\omega^2}$.

Substituting the constants into the general solution,

$$\boxed{x(t) = 7\cos\omega t - \frac{t\cos\omega t}{2\omega} + \frac{\sin\omega t}{2\omega^2}}$$

Check of the resonant solution by the limit $\alpha \to \omega$ of the forced part of item (vi), the fraction, which does not depend on the initial data (L'Hopital's rule, differentiating numerator and denominator with respect to α):

$$\lim_{\alpha\to\omega} \frac{\sin\alpha t - \frac{\alpha}{\omega}\sin\omega t}{\omega^2 - \alpha^2} = \lim_{\alpha\to\omega} \frac{t\cos\alpha t - \frac{1}{\omega}\sin\omega t}{-2\alpha} = \frac{t\cos\omega t - \frac{1}{\omega}\sin\omega t}{-2\omega} = -\frac{t\cos\omega t}{2\omega} + \frac{\sin\omega t}{2\omega^2}$$

which is exactly the zero-data (forced) part of the resonant solution just found, i.e. everything except $7\cos\omega t$: the resonant solution is the limit of the non-resonant ones.

(viii) $\ddot{x} + 2\dot{x} + 10x = e^{-t}$**, with** $x(0) = 8,\ \dot{x}(0) = 0$

The homogeneous equation $\ddot{x} + 2\dot{x} + 10x = 0$ has constant coefficients, so we try $x = e^{kt}$ (Robinson Sec. 12.1 and 16.1): by the chain rule each derivative multiplies by k, and dividing by $e^{kt} \neq 0$ leaves the characteristic equation

$$P(k) = k^2 + 2k + 10 = 0$$

Its discriminant is $\Delta = b^2 - 4ac = (2)^2 - 4(1)(10) = -36$.

Factorising, $P(k) = (k + 1 - 3i)\,(k + 1 + 3i)$, so the roots are

$$k = -1 - 3i, \qquad k = -1 + 3i$$

A real root k of multiplicity m gives $t^j e^{kt}$ for $j = 0, \ldots, m-1$; a complex pair $\rho \pm i\omega$ gives $t^j e^{\rho t}\cos\omega t$ and $t^j e^{\rho t}\sin\omega t$ (Euler's formula, Robinson Sec. 12.3). The fundamental set is $e^{-t}\cos 3t$, $e^{-t}\sin 3t$, so the complementary function is

$$x_h(t) = c_1 e^{-t}\cos 3t + c_2 e^{-t}\sin 3t$$

The forcing e^{-t} is an exponential with exponent $s = -1$; we test it in the characteristic polynomial: $P(-1) = 9$.

Since $P(s) \neq 0$, no term of the standard guess solves the homogeneous equation, so the standard guess is used as it stands.

Guess for the particular integral, with undetermined coefficients:

$$x_p(t) = K_1 e^{-t}$$

Differentiate term by term (product rule on $t^j e^{\sigma t}$, chain rule on $\cos \omega t$ and $\sin \omega t$):

$$x_p'(t) = -K_1 e^{-t}$$
$$x_p''(t) = K_1 e^{-t}$$

Substitute into $L[x_p] = x_p'' + 2x_p' + 10x_p$ and collect the terms that multiply the same function:

$$L[x_p] = 9K_1\, e^{-t}$$

This must equal the forcing $f(t) = e^{-t}$ for every t. The functions on the left are linearly independent, so the coefficient of each must match the coefficient of the same function in f:

$$e^{-t}: \quad 9K_1 = 1$$

The linear system is

$$9K_1 = 1 \quad (1)$$

$9K_1 = 1$ contains one unknown only, so $K_1 = \frac{1}{9}$.

Hence $K_1 = \frac{1}{9}$.

Therefore

$$x_p(t) = \frac{e^{-t}}{9}$$

The general solution is the complementary function plus the particular integral:

$$x(t) = x_h(t) + x_p(t) = c_1 e^{-t}\cos 3t + c_2 e^{-t}\sin 3t + \frac{e^{-t}}{9}$$

Initial conditions. Differentiate the general solution (product and chain rules):

$$x'(t) = -3c_1 e^{-t}\sin 3t - c_1 e^{-t}\cos 3t - c_2 e^{-t}\sin 3t + 3c_2 e^{-t}\cos 3t - \frac{e^{-t}}{9}$$

Evaluate at $t = 0$, using $e^0 = 1$, $\cos 0 = 1$ and $\sin 0 = 0$:

$$x(0) = c_1 + \frac{1}{9} = 8$$
$$x'(0) = -c_1 + 3c_2 - \frac{1}{9} = 0$$

The linear system is

$$c_1 + \frac{1}{9} = 8 \quad (1)$$
$$-c_1 + 3c_2 - \frac{1}{9} = 0 \quad (2)$$

$c_1 + \frac{1}{9} = 8$ contains one unknown only, so $c_1 = \frac{71}{9}$.

Substitute in $-c_1 + 3c_2 - \frac{1}{9} = 0$: $3c_2 - 8 = 0$.

$3c_2 - 8 = 0$ contains one unknown only, so $c_2 = \frac{8}{3}$.

Hence $c_1 = \frac{71}{9}$, $c_2 = \frac{8}{3}$.

Substituting the constants into the general solution,

$$\boxed{x(t) = \frac{8e^{-t}\sin 3t}{3} + \frac{71e^{-t}\cos 3t}{9} + \frac{e^{-t}}{9}}$$

(ix) $\ddot{x} + 2\dot{x} + 10x = e^{-t}\cos 3t$**, with** $x(0) = 9$**,** $\dot{x}(0) = 0$

The homogeneous equation $\ddot{x} + 2\dot{x} + 10x = 0$ has constant coefficients, so we try $x = e^{kt}$ (Robinson Sec. 12.1 and 16.1): by the chain rule each derivative multiplies by k, and dividing by $e^{kt} \neq 0$ leaves the characteristic equation

$$P(k) = k^2 + 2k + 10 = 0$$

Its discriminant is $\Delta = b^2 - 4ac = (2)^2 - 4(1)(10) = -36$.

Factorising, $P(k) = (k + 1 - 3i)\,(k + 1 + 3i)$, so the roots are

$$k = -1 - 3i, \qquad k = -1 + 3i$$

A real root k of multiplicity m gives $t^j e^{kt}$ for $j = 0, \dots, m-1$; a complex pair $\rho \pm i\omega$ gives $t^j e^{\rho t}\cos \omega t$ and $t^j e^{\rho t}\sin \omega t$ (Euler's formula, Robinson Sec. 12.3). The fundamental set is $e^{-t}\cos 3t$, $e^{-t}\sin 3t$, so the complementary function is

$$x_h(t) = c_1 e^{-t}\cos 3t + c_2 e^{-t}\sin 3t$$

The forcing $e^{-t}\cos 3t$ is a sinusoidal term times an exponential: it is the real or imaginary part of $e^{(-1+3i)t}$, so we test $s = -1 + 3i$ in the characteristic polynomial: $P(-1+3i) = 0$.

Since $P(s) = 0$ and s is a root of multiplicity 1, the standard guess would contain terms of the complementary function. By the rule of thumb (Robinson Sec. 14.5, p. 139) the guess is multiplied by t^1.

Guess for the particular integral, with undetermined coefficients:

$$x_p(t) = K_1 t e^{-t}\cos 3t + K_2 t e^{-t}\sin 3t$$

Differentiate term by term (product rule on $t^j e^{\sigma t}$, chain rule on $\cos \omega t$ and $\sin \omega t$):

$$\begin{aligned} x_p'(t) &= -3K_1 t e^{-t}\sin 3t - K_1 t e^{-t}\cos 3t + K_1 e^{-t}\cos 3t - K_2 t e^{-t}\sin 3t + 3K_2 t e^{-t}\cos 3t + K_2 e^{-t}\sin 3t \\ x_p''(t) &= 6K_1 t e^{-t}\sin 3t - 8K_1 t e^{-t}\cos 3t - 6K_1 e^{-t}\sin 3t - 2K_1 e^{-t}\cos 3t - 8K_2 t e^{-t}\sin 3t - 6K_2 t e^{-t}\cos 3t - 2K_2 e^{-t}\sin 3t + 6K_2 e^{-t}\cos 3t \end{aligned}$$

Substitute into $L[x_p] = x_p'' + 2x_p' + 10x_p$ and collect the terms that multiply the same function:

$$L[x_p] = 6K_2\, e^{-t}\cos 3t - 6K_1\, e^{-t}\sin 3t$$

This must equal the forcing $f(t) = e^{-t}\cos 3t$ for every t. The functions on the left are linearly independent, so the coefficient of each must match the coefficient of the same function in f:

$$\begin{aligned} e^{-t}\cos 3t: &\quad 6K_2 = 1 \\ e^{-t}\sin 3t: &\quad -6K_1 = 0 \end{aligned}$$

The linear system is

$$\begin{aligned} 6K_2 &= 1 \quad (1) \\ -6K_1 &= 0 \quad (2) \end{aligned}$$

$6K_2 = 1$ contains one unknown only, so $K_2 = \frac{1}{6}$.

$-6K_1 = 0$ contains one unknown only, so $K_1 = 0$.

Hence $K_1 = 0, K_2 = \frac{1}{6}$.

Therefore

$$x_p(t) = \frac{te^{-t}\sin 3t}{6}$$

The general solution is the complementary function plus the particular integral:

$$x(t) = x_h(t) + x_p(t) = c_1 e^{-t}\cos 3t + c_2 e^{-t}\sin 3t + \frac{te^{-t}\sin 3t}{6}$$

Initial conditions. Differentiate the general solution (product and chain rules):

$$x'(t) = -3c_1 e^{-t}\sin 3t - c_1 e^{-t}\cos 3t - c_2 e^{-t}\sin 3t + 3c_2 e^{-t}\cos 3t - \frac{te^{-t}\sin 3t}{6} + \frac{te^{-t}\cos 3t}{2} + \frac{e^{-t}\sin 3t}{6}$$

Evaluate at $t = 0$, using $e^0 = 1$, $\cos 0 = 1$ and $\sin 0 = 0$:

$$\begin{aligned} x(0) &= c_1 = 9 \\ x'(0) &= -c_1 + 3c_2 = 0 \end{aligned}$$

The linear system is

$$\begin{aligned} c_1 &= 9 \quad (1) \\ -c_1 + 3c_2 &= 0 \quad (2) \end{aligned}$$

$c_1 = 9$ contains one unknown only, so $c_1 = 9$.

Substitute in $-c_1 + 3c_2 = 0$: $3c_2 - 9 = 0$.

$3c_2 - 9 = 0$ contains one unknown only, so $c_2 = 3$.

Hence $c_1 = 9, c_2 = 3$.

Substituting the constants into the general solution,

$$\boxed{x(t) = \frac{te^{-t}\sin 3t}{6} + 3e^{-t}\sin 3t + 9e^{-t}\cos 3t}$$

(x) $\ddot{x} + 6\dot{x} + 10x = e^{-3t}\cos t$**, with** $x(0) = 10,\ \dot{x}(0) = 0$

The homogeneous equation $\ddot{x} + 6\dot{x} + 10x = 0$ has constant coefficients, so we try $x = e^{kt}$ (Robinson Sec. 12.1 and 16.1): by the chain rule each derivative multiplies by k, and dividing by $e^{kt} \neq 0$ leaves the characteristic equation

$$P(k) = k^2 + 6k + 10 = 0$$

Its discriminant is $\Delta = b^2 - 4ac = (6)^2 - 4(1)(10) = -4$.

Factorising, $P(k) = (k + 3 - i)\,(k + 3 + i)$, so the roots are

$$k = -3 - i, \qquad k = -3 + i$$

A real root k of multiplicity m gives $t^j e^{kt}$ for $j = 0, \ldots, m-1$; a complex pair $\rho \pm i\omega$ gives $t^j e^{\rho t}\cos\omega t$ and $t^j e^{\rho t}\sin\omega t$ (Euler's formula, Robinson Sec. 12.3). The fundamental set is $e^{-3t}\cos t$, $e^{-3t}\sin t$, so the complementary function is

$$x_h(t) = c_1 e^{-3t}\cos t + c_2 e^{-3t}\sin t$$

The forcing $e^{-3t}\cos t$ is a sinusoidal term times an exponential: it is the real or imaginary part of $e^{(-3+i)t}$, so we test $s = -3 + i$ in the characteristic polynomial: $P(-3 + i) = 0$.

Since $P(s) = 0$ and s is a root of multiplicity 1, the standard guess would contain terms of the complementary function. By the rule of thumb (Robinson Sec. 14.5, p. 139) the guess is multiplied by t^1.

Guess for the particular integral, with undetermined coefficients:

$$x_p(t) = K_1 te^{-3t}\cos t + K_2 te^{-3t}\sin t$$

Differentiate term by term (product rule on $t^j e^{\sigma t}$, chain rule on $\cos\omega t$ and $\sin\omega t$):

$$x_p'(t) = -K_1 t e^{-3t}\sin t - 3K_1 t e^{-3t}\cos t + K_1 e^{-3t}\cos t - 3K_2 t e^{-3t}\sin t + K_2 t e^{-3t}\cos t + K_2 e^{-3t}\sin t$$
$$x_p''(t) = 6K_1 t e^{-3t}\sin t + 8K_1 t e^{-3t}\cos t - 2K_1 e^{-3t}\sin t - 6K_1 e^{-3t}\cos t + 8K_2 t e^{-3t}\sin t - 6K_2 t e^{-3t}\cos t - 6K_2 e^{-3t}\sin t + 2K_2 e^{-3t}\cos t$$

Substitute into $L[x_p] = x_p'' + 6x_p' + 10x_p$ and collect the terms that multiply the same function:

$$L[x_p] = 2K_2\, e^{-3t}\cos t - 2K_1\, e^{-3t}\sin t$$

This must equal the forcing $f(t) = e^{-3t}\cos t$ for every t. The functions on the left are linearly independent, so the coefficient of each must match the coefficient of the same function in f:

$$\begin{aligned} e^{-3t}\cos t: &\quad 2K_2 = 1 \\ e^{-3t}\sin t: &\quad -2K_1 = 0 \end{aligned}$$

The linear system is

$$\begin{aligned} 2K_2 &= 1 \quad (1) \\ -2K_1 &= 0 \quad (2) \end{aligned}$$

$2K_2 = 1$ contains one unknown only, so $K_2 = \frac{1}{2}$.

$-2K_1 = 0$ contains one unknown only, so $K_1 = 0$.

Hence $K_1 = 0, K_2 = \frac{1}{2}$.

Therefore

$$x_p(t) = \frac{te^{-3t}\sin t}{2}$$

The general solution is the complementary function plus the particular integral:

$$x(t) = x_h(t) + x_p(t) = c_1 e^{-3t}\cos t + c_2 e^{-3t}\sin t + \frac{te^{-3t}\sin t}{2}$$

Initial conditions. Differentiate the general solution (product and chain rules):

$$x'(t) = -c_1 e^{-3t}\sin t - 3c_1 e^{-3t}\cos t - 3c_2 e^{-3t}\sin t + c_2 e^{-3t}\cos t - \frac{3te^{-3t}\sin t}{2} + \frac{te^{-3t}\cos t}{2} + \frac{e^{-3t}\sin t}{2}$$

Evaluate at $t = 0$, using $e^0 = 1$, $\cos 0 = 1$ and $\sin 0 = 0$:

$$x(0) = c_1 = 10$$
$$x'(0) = -3c_1 + c_2 = 0$$

The linear system is

$$c_1 = 10 \quad (1)$$
$$-3c_1 + c_2 = 0 \quad (2)$$

$c_1 = 10$ contains one unknown only, so $c_1 = 10$.

Substitute in $-3c_1 + c_2 = 0$: $c_2 - 30 = 0$.

$c_2 - 30 = 0$ contains one unknown only, so $c_2 = 30$.

Hence $c_1 = 10$, $c_2 = 30$.

Substituting the constants into the general solution,

$$\boxed{x(t) = \frac{te^{-3t}\sin t}{2} + 30e^{-3t}\sin t + 10e^{-3t}\cos t}$$

(xi) $\ddot{x} + 4\dot{x} + 4x = e^{2t}$**, with** $x(0) = 11$**,** $\dot{x}(0) = 0$

The homogeneous equation $\ddot{x} + 4\dot{x} + 4x = 0$ has constant coefficients, so we try $x = e^{kt}$ (Robinson Sec. 12.1 and 16.1): by the chain rule each derivative multiplies by k, and dividing by $e^{kt} \neq 0$ leaves the characteristic equation

$$P(k) = k^2 + 4k + 4 = 0$$

Its discriminant is $\Delta = b^2 - 4ac = (4)^2 - 4(1)(4) = 0$.

Factorising, $P(k) = (k+2)^2$, so the roots are

$$k = -2 \text{ (multiplicity 2)}$$

A real root k of multiplicity m gives $t^j e^{kt}$ for $j = 0, \ldots, m-1$; a complex pair $\rho \pm i\omega$ gives $t^j e^{\rho t}\cos\omega t$ and $t^j e^{\rho t}\sin\omega t$ (Euler's formula, Robinson Sec. 12.3). The fundamental set is e^{-2t}, te^{-2t}, so the complementary function is

$$x_h(t) = c_1 e^{-2t} + c_2 t e^{-2t}$$

The forcing e^{2t} is an exponential with exponent $s = 2$; we test it in the characteristic polynomial: $P(2) = 16$.

Since $P(s) \neq 0$, no term of the standard guess solves the homogeneous equation, so the standard guess is used as it stands.

Guess for the particular integral, with undetermined coefficients:

$$x_p(t) = K_1 e^{2t}$$

Differentiate term by term (product rule on $t^j e^{\sigma t}$, chain rule on $\cos \omega t$ and $\sin \omega t$):

$$x_p'(t) = 2K_1 e^{2t}$$
$$x_p''(t) = 4K_1 e^{2t}$$

Substitute into $L[x_p] = x_p'' + 4x_p' + 4x_p$ and collect the terms that multiply the same function:

$$L[x_p] = 16K_1\, e^{2t}$$

This must equal the forcing $f(t) = e^{2t}$ for every t. The functions on the left are linearly independent, so the coefficient of each must match the coefficient of the same function in f:

$$e^{2t}: \quad 16K_1 = 1$$

The linear system is

$$16K_1 = 1 \quad (1)$$

$16K_1 = 1$ contains one unknown only, so $K_1 = \frac{1}{16}$.

Hence $K_1 = \frac{1}{16}$.

Therefore

$$x_p(t) = \frac{e^{2t}}{16}$$

The general solution is the complementary function plus the particular integral:

$$x(t) = x_h(t) + x_p(t) = c_1 e^{-2t} + c_2 t e^{-2t} + \frac{e^{2t}}{16}$$

Initial conditions. Differentiate the general solution (product and chain rules):

$$x'(t) = -2c_1 e^{-2t} - 2c_2 t e^{-2t} + c_2 e^{-2t} + \frac{e^{2t}}{8}$$

Evaluate at $t = 0$, using $e^0 = 1$, $\cos 0 = 1$ and $\sin 0 = 0$:

$$x(0) = c_1 + \frac{1}{16} = 11$$
$$x'(0) = -2c_1 + c_2 + \frac{1}{8} = 0$$

The linear system is

$$c_1 + \frac{1}{16} = 11 \quad (1)$$
$$-2c_1 + c_2 + \frac{1}{8} = 0 \quad (2)$$

$c_1 + \frac{1}{16} = 11$ contains one unknown only, so $c_1 = \frac{175}{16}$.

Substitute in $-2c_1 + c_2 + \frac{1}{8} = 0$: $c_2 - \frac{87}{4} = 0$.

$c_2 - \frac{87}{4} = 0$ contains one unknown only, so $c_2 = \frac{87}{4}$.

Hence $c_1 = \frac{175}{16}, c_2 = \frac{87}{4}$.

Substituting the constants into the general solution,

$$\boxed{x(t) = \frac{87te^{-2t}}{4} + \frac{e^{2t}}{16} + \frac{175e^{-2t}}{16}}$$

PHASE 4: INTERPRETATION, LONG-TIME BEHAVIOUR AND CHECKS

Summary of the eleven solutions of the IVPs (item (n) has $x(0) = n, \dot{x}(0) = 0$):

Item	**Solution** $x(t)$
(i)	$-\frac{t^2}{4} + \frac{9e^{2t}}{16} - \frac{1}{8} + \frac{9e^{-2t}}{16}$
(ii)	$-\frac{t^3}{12} - \frac{t^2}{16} - \frac{t}{32} + \frac{e^{4t}}{128} + \frac{255}{128}$
(iii)	$\frac{5e^{t}}{2} - \frac{3e^{-t}}{2} + 2e^{-2t}$
(iv)	$\frac{te^{t}}{3} + \frac{23e^{t}}{9} + \frac{13e^{-2t}}{9}$
(v)	$\frac{t^2e^{-t}}{2} + 5te^{-t} + 5e^{-t}$
(vi)	$6\cos\omega t + \dfrac{\sin\alpha t - (\alpha/\omega)\sin\omega t}{\omega^2 - \alpha^2}$
(vii)	$7\cos\omega t - \frac{t\cos\omega t}{2\omega} + \frac{\sin\omega t}{2\omega^2}$
(viii)	$\frac{8e^{-t}\sin 3t}{3} + \frac{71e^{-t}\cos 3t}{9} + \frac{e^{-t}}{9}$
(ix)	$\frac{te^{-t}\sin 3t}{6} + 3e^{-t}\sin 3t + 9e^{-t}\cos 3t$
(x)	$\frac{te^{-3t}\sin t}{2} + 30e^{-3t}\sin t + 10e^{-3t}\cos t$
(xi)	$\frac{87te^{-2t}}{4} + \frac{e^{2t}}{16} + \frac{175e^{-2t}}{16}$

Behaviour as $t \to \infty$

- (i): all terms of the complementary function except the growing one are small: $x \sim \frac{9}{16}e^{2t}$, unbounded.
- (ii): the term $\frac{1}{128}e^{4t}$ is tiny at first but eventually dominates the cubic polynomial: unbounded.
- (iii): $x \sim \frac{5}{2}e^{t}$ (the forcing $3e^{-t}$ and the mode e^{-2t} decay): unbounded.
- (iv): resonance with the root $k = 1$: $x \sim (\frac{t}{3} + \frac{23}{9})e^{t}$, faster than a pure exponential.
- (v): every term carries e^{-t}: $x \to 0$, although the resonant term $\frac{1}{2}t^2e^{-t}$ first grows (critical damping, resonant forcing).
- (vi): bounded for all t (a sum of two sinusoids, periodic exactly when α/ω is rational); no decay; beat-like modulation appears when $\alpha \approx \omega$.
- (vii): resonance: the amplitude of $-\dfrac{t\cos\omega t}{2\omega}$ grows linearly without bound.
- (viii): all terms decay like e^{-t}: $x \to 0$; the non-oscillatory forced part $\frac{1}{9}e^{-t}$ is not resonant.
- (ix): resonance with the damped oscillation: the envelope $\frac{t}{6}e^{-t}$ peaks at $t = 1$ (value $\frac{1}{6e} \approx 0.061$) and then decays; $x \to 0$.
- (x): resonance with $e^{-3t}\cos t$: the envelope $\frac{t}{2}e^{-3t}$ peaks at $t = \frac{1}{3}$ and decays; $x \to 0$.
- (xi): $x \sim \frac{1}{16}e^{2t}$: the forced exponential beats the decaying modes e^{-2t}, te^{-2t}: unbounded.

Checks

- Every solution was substituted back into its equation and into both initial conditions symbolically (script `scripts/verify_topic3_topic4.py`): the residual $L[x] - f$ is identically zero and $x(0) = n$, $\dot{x}(0) = 0$.

- Dimensions: if x is a length and t a time, every term of $\ddot{x} + b\dot{x} + cx = f$ has units of length/time^2, so b has units s^{-1}, c and ω^2 units s^{-2} and f units m/s^2 if the equation is read dimensionally. The sheet itself uses dimensionless variables (forcings such as t^2 or e^{-t} carry no units), so these units only serve to check the structure of the formulas. In (vi) the combination α/ω is dimensionless and $1/(\omega^2 - \alpha^2)$ has units s^2, so x comes out in metres.
- Resonance diagnostic: the items with a factor t in the particular integral are exactly those where the exponent of f is a root of P (listed in Phase 2).
- Beats versus resonance: as $\alpha \to \omega$, (vi) tends to (vii) (L'Hopital check above); the beat amplitude $\propto 1/(\omega^2 - \alpha^2)$ blows up exactly where the resonant solution replaces it.

EXERCISE 2

A Particular Integral by Superposition

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Superposition • Exponential Forcing

STATEMENT

> Find a particular solution for $\ddot{x} + \dot{x} - 2x = 12e^{-t} - 6e^{t}$. You may find useful parts (iii) and (iv) of the previous exercise.

PHASE 1: SETUP

The equation is $L[x] = \ddot{x} + \dot{x} - 2x = f$ with $f = 12e^{-t} - 6e^{t}$, the same operator L as items (iii) and (iv) of Exercise 1. L is linear (Robinson Eq. 11.11), so if $L[x_a] = f_a$ and $L[x_b] = f_b$ then $L[\alpha x_a + \beta x_b] = \alpha f_a + \beta f_b$ (Robinson Eq. 14.7). Hypothesis: nothing more is needed, since any single solution of $L[x] = f$ is a particular integral.

PHASE 2: FORMULATION

Rewrite the forcing in terms of the two forcings of Exercise 1:

$$f = 12e^{-t} - 6e^{t} = 4\underbrace{\left(3e^{-t}\right)}_{f_a \text{ (item iii)}} - 6\underbrace{\left(e^{t}\right)}_{f_b \text{ (item iv)}}$$

Item (iii) found x_a with $L[x_a] = 3e^{-t}$ and item (iv) found x_b with $L[x_b] = e^{t}$. Then $x_p = 4x_a - 6x_b$.

PHASE 3: DERIVATION

The two particular integrals of Exercise 1

$P(k) = k^2 + k - 2 = (k-1)(k+2)$ has roots 1 and -2. For $3e^{-t}$ the exponent is $s = -1$, $P(-1) = 1 - 1 - 2 = -2 \neq 0$, so $x_a = Ke^{-t}$ and, since $\dot{x}_a = -Ke^{-t}$, $\ddot{x}_a = Ke^{-t}$: $L[x_a] = (K - K - 2K)e^{-t} = -2Ke^{-t} = 3e^{-t}$, hence $K = -\frac{3}{2}$:

$$x_a(t) = -\tfrac{3}{2}e^{-t}$$

For e^t the exponent is $s = 1$, a simple root ($P(1) = 1 + 1 - 2 = 0$), so the guess is $x_b = Kte^t$. By the product rule $\dot{x}_b = K(1+t)e^t$ and $\ddot{x}_b = K(2+t)e^t$, so

$$L[x_b] = K\left[(2+t) + (1+t) - 2t\right]e^t = 3Ke^t = e^t \implies K = \tfrac{1}{3}, \qquad x_b(t) = \tfrac{1}{3}\,t\,e^t$$

Superposition

$$x_p = 4x_a - 6x_b = 4\left(-\tfrac{3}{2}e^{-t}\right) - 6\left(\tfrac{1}{3}te^t\right) = -2te^t - 6e^{-t}$$

Direct verification by substitution

$$\dot{x}_p = -2te^t - 2e^t + 6e^{-t}$$
$$\ddot{x}_p = -2te^t - 4e^t - 6e^{-t}$$

$$\ddot{x}_p + \dot{x}_p - 2x_p = \left(-2te^t - 4e^t - 6e^{-t}\right) + \left(-2te^t - 2e^t + 6e^{-t}\right) - 2\left(-2te^t - 6e^{-t}\right) = -6e^t + 12e^{-t}$$

This is $12e^{-t} - 6e^t$, as required.

PHASE 4: INTERPRETATION AND CHECKS

RESULT

A particular solution is $x_p(t) = -2te^t - 6e^{-t}$.

The general solution is $x(t) = c_1e^t + c_2e^{-2t} - 6e^{-t} - 2te^t$ (the roots 1 and -2 of P give the complementary function).

- The term $-2te^t$ carries the extra factor t because e^t solves the homogeneous equation (resonance); the term $-6e^{-t}$ does not.
- Particular integrals are not unique: adding any multiple of e^t or e^{-2t} gives another one; the answer above is the one with no such terms.
- Dimensions: with $[t] = \mathrm{s}$, each term of $L[x]$ has the units of f; both sides are exponentials of a dimensionless argument $t/(1\ \mathrm{s})$, so the check is purely numerical.

EXERCISE 3

Harmonic Oscillator under Periodic Forcing: Beats and Resonance

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Forced Oscillations • Resonance • Beats

STATEMENT

Suppose that $x(t)$ denotes the distance of some system from its equilibrium position, such that it satisfies the differential equation

$$\ddot{x} + \omega^2 x = a \cos \alpha t,$$

where the second term on the left hand side accounts for harmonic oscillations and the right hand side is an external forcing that is periodic in time. Discuss the form of the general solution in terms of the natural frequency of oscillation ω and the frequency α of the external force.

PHASE 1: SETUP AND CLASSIFICATION

Hypotheses: $\omega > 0$ is the natural angular frequency (rad/s), $\alpha \geq 0$ the forcing frequency and a the forcing amplitude per unit mass (m/s^2). The equation is linear, second-order, with constant coefficients and a sinusoidal forcing. Its characteristic polynomial $P(k) = k^2 + \omega^2$ has the imaginary roots $k = \pm i\omega$, so the complementary function is the simple harmonic motion $x_h = c_1 \cos \omega t + c_2 \sin \omega t$ (Topic 3, Sec. 4.2).

The forcing $a \cos \alpha t$ has exponent $s = \pm i\alpha$. The whole discussion hinges on whether s is a root of P, that is, on whether $\alpha = \omega$ (**resonance**) or $\alpha \neq \omega$. Both cases are solved below with the method of undetermined coefficients, then compared.

PHASE 2: METHOD

General solution $x = x_h + x_p$ (Robinson Sec. 14.1). For x_p use the guess table of Topic 3, Sec. 5: $K_1 \cos \alpha t + K_2 \sin \alpha t$ when $P(i\alpha) \neq 0$ and $t(K_1 \cos \omega t + K_2 \sin \omega t)$ when $\alpha = \omega$.

PHASE 3: DERIVATION

Case A: $\alpha \neq \omega$ (non-resonant forcing)

The homogeneous equation $\ddot{x} + \omega^2 x = 0$ has constant coefficients, so we try $x = e^{kt}$ (Robinson Sec. 12.1 and 16.1): by the chain rule each derivative multiplies by k, and dividing by $e^{kt} \neq 0$ leaves the characteristic equation

$$P(k) = k^2 + \omega^2 = 0$$

Its discriminant is $\Delta = b^2 - 4ac = (0)^2 - 4(1)(\omega^2) = -4\omega^2$.

Factorising, $P(k) = (k - i\omega)\,(k + i\omega)$, so the roots are

$$k = -i\omega, \qquad k = i\omega$$

A real root k of multiplicity m gives $t^j e^{kt}$ for $j = 0, \ldots, m-1$; a complex pair $\rho \pm i\omega$ gives $t^j e^{\rho t}\cos\omega t$ and $t^j e^{\rho t}\sin\omega t$ (Euler's formula, Robinson Sec. 12.3). The fundamental set is $\cos\omega t$, $\sin\omega t$, so the complementary function is

$$x_h(t) = c_1 \cos\omega t + c_2 \sin\omega t$$

The forcing $a\cos\alpha t$ is a sinusoidal term: it is the real or imaginary part of $e^{(i\alpha)t}$, so we test $s = i\alpha$ in the characteristic polynomial: $P(i\alpha) = -\alpha^2 + \omega^2$.

Since $P(s) \neq 0$, no term of the standard guess solves the homogeneous equation, so the standard guess is used as it stands.

Guess for the particular integral, with undetermined coefficients:

$$x_p(t) = K_1 \cos\alpha t + K_2 \sin\alpha t$$

Differentiate term by term (product rule on $t^j e^{\sigma t}$, chain rule on $\cos\omega t$ and $\sin\omega t$):

$$\begin{aligned} x_p'(t) &= -K_1\alpha \sin\alpha t + K_2\alpha \cos\alpha t \\ x_p''(t) &= -K_1\alpha^2 \cos\alpha t - K_2\alpha^2 \sin\alpha t \end{aligned}$$

Substitute into $L[x_p] = x_p'' + \omega^2 x_p$ and collect the terms that multiply the same function:

$$L[x_p] = \left(-K_1\alpha^2 + K_1\omega^2\right)\,\cos\alpha\,t + \left(-K_2\alpha^2 + K_2\omega^2\right)\,\sin\alpha\,t$$

This must equal the forcing $f(t) = a\cos\alpha t$ for every t. The functions on the left are linearly independent, so the coefficient of each must match the coefficient of the same function in f:

$$\begin{aligned}\cos\alpha\,t: &\quad -K_1\alpha^2 + K_1\omega^2 = a\\ \sin\alpha\,t: &\quad -K_2\alpha^2 + K_2\omega^2 = 0\end{aligned}$$

The linear system is

$$\begin{aligned}-K_1\alpha^2 + K_1\omega^2 &= a \quad (1)\\ -K_2\alpha^2 + K_2\omega^2 &= 0 \quad (2)\end{aligned}$$

$-K_1\alpha^2 + K_1\omega^2 = a$ contains one unknown only, so $K_1 = -\frac{a}{\alpha^2-\omega^2}$.

$-K_2\alpha^2 + K_2\omega^2 = 0$ contains one unknown only, so $K_2 = 0$.

Hence $K_1 = -\frac{a}{\alpha^2-\omega^2}, K_2 = 0$.

Therefore

$$x_p(t) = -\frac{a\cos\alpha t}{\alpha^2 - \omega^2}$$

The general solution is the complementary function plus the particular integral:

$$x(t) = x_h(t) + x_p(t) = -\frac{a\cos\alpha t}{\alpha^2 - \omega^2} + c_1\cos\omega t + c_2\sin\omega t$$

Remarks on this case. The guess contained a sine term, but matching showed $K_2 = 0$: with no $\dot{x}$ term in the equation, cosines are only turned into cosines. The general solution is

$$x(t) = c_1\cos\omega t + c_2\sin\omega t + \frac{a}{\omega^2 - \alpha^2}\cos\alpha t = M\cos(\omega t - \phi) + \frac{a}{\omega^2 - \alpha^2}\cos\alpha t$$

(the last form uses the amplitude-phase rewriting $c_1\cos\omega t + c_2\sin\omega t = M\cos(\omega t - \phi)$ with $M = \sqrt{c_1^2 + c_2^2}$, Topic 3, Eq. 13.3). The solution is a superposition of two sinusoids with different frequencies, so it is **bounded** for every t; it is periodic only if ω/α is rational. The forced amplitude $|a/(\omega^2 - \alpha^2)|$ is large when $\alpha \approx \omega$.

Starting from rest: beats

With $x(0) = 0, \dot{x}(0) = 0$: $x(0) = c_1 + \frac{a}{\omega^2 - \alpha^2} = 0$ gives $c_1 = -\frac{a}{\omega^2 - \alpha^2}$, and $\dot{x}(0) = \omega c_2 = 0$ gives $c_2 = 0$ (the forced term $\propto \cos \alpha t$ has zero slope at 0). Hence

$$x(t) = \frac{a}{\omega^2 - \alpha^2} \left(\cos \alpha t - \cos \omega t \right)$$

Use $\cos A - \cos B = -2 \sin \frac{A+B}{2} \sin \frac{A-B}{2}$ with $A = \alpha t$, $B = \omega t$, noting that $\sin \frac{\alpha - \omega}{2} t = -\sin \frac{\omega - \alpha}{2} t$:

$$x(t) = \frac{2a}{\omega^2 - \alpha^2} \, \sin \frac{(\omega + \alpha)t}{2} \, \sin \frac{(\omega - \alpha)t}{2}$$

This is a fast oscillation at the mean frequency $(\omega + \alpha)/2$ whose amplitude is modulated by the slow envelope $\pm \frac{2a}{|\omega^2 - \alpha^2|} \sin \frac{|\omega - \alpha| t}{2}$: **beats**. The envelope vanishes every $2\pi / |\omega - \alpha|$ (the beat period; the signed factor $\sin \frac{(\omega - \alpha)t}{2}$ itself has period $4\pi / |\omega - \alpha|$), long when $\alpha \approx \omega$, and its peak value $\frac{2a}{|\omega^2 - \alpha^2|}$ grows without bound as $\alpha \to \omega$.

Case B: $\alpha = \omega$ (resonance)

The homogeneous equation $\ddot{x} + \omega^2 x = 0$ has constant coefficients, so we try $x = e^{kt}$ (Robinson Sec. 12.1 and 16.1): by the chain rule each derivative multiplies by k, and dividing by $e^{kt} \neq 0$ leaves the characteristic equation

$$P(k) = k^2 + \omega^2 = 0$$

Its discriminant is $\Delta = b^2 - 4ac = (0)^2 - 4(1)(\omega^2) = -4\omega^2$.

Factorising, $P(k) = (k - i\omega)\,(k + i\omega)$, so the roots are

$$k = -i\omega, \qquad k = i\omega$$

A real root k of multiplicity m gives $t^j e^{kt}$ for $j = 0, \ldots, m-1$; a complex pair $\rho \pm i\omega$ gives $t^j e^{\rho t} \cos \omega t$ and $t^j e^{\rho t} \sin \omega t$ (Euler's formula, Robinson Sec. 12.3). The fundamental set is $\cos \omega t$, $\sin \omega t$, so the complementary function is

$$x_h(t) = c_1 \cos \omega t + c_2 \sin \omega t$$

The forcing $a \cos \omega t$ is a sinusoidal term: it is the real or imaginary part of $e^{(i\omega)t}$, so we test $s = i\omega$ in the characteristic polynomial: $P(i\omega) = 0$.

Since $P(s) = 0$ and s is a root of multiplicity 1, the standard guess would contain terms of the complementary function. By the rule of thumb (Robinson Sec. 14.5, p. 139) the guess is multiplied by t^1.

Guess for the particular integral, with undetermined coefficients:

$$x_p(t) = K_1 t \cos \omega t + K_2 t \sin \omega t$$

Differentiate term by term (product rule on $t^j e^{\sigma t}$, chain rule on $\cos \omega t$ and $\sin \omega t$):

$$\begin{aligned} x_p'(t) &= -K_1 \omega t \sin \omega t + K_1 \cos \omega t + K_2 \omega t \cos \omega t + K_2 \sin \omega t \\ x_p''(t) &= -K_1 \omega^2 t \cos \omega t - 2K_1 \omega \sin \omega t - K_2 \omega^2 t \sin \omega t + 2K_2 \omega \cos \omega t \end{aligned}$$

Substitute into $L[x_p] = x_p'' + \omega^2 x_p$ and collect the terms that multiply the same function:

$$L[x_p] = 2K_2 \omega \, \cos \omega \, t - 2K_1 \omega \, \sin \omega \, t$$

This must equal the forcing $f(t) = a \cos \omega t$ for every t. The functions on the left are linearly independent, so the coefficient of each must match the coefficient of the same function in f:

$$\begin{aligned} \cos \omega \, t: &\quad 2K_2 \omega = a \\ \sin \omega \, t: &\quad -2K_1 \omega = 0 \end{aligned}$$

The linear system is

$$\begin{aligned} 2K_2 \omega &= a \quad (1) \\ -2K_1 \omega &= 0 \quad (2) \end{aligned}$$

$2K_2 \omega = a$ contains one unknown only, so $K_2 = \frac{a}{2\omega}$.

$-2K_1 \omega = 0$ contains one unknown only, so $K_1 = 0$.

Hence $K_1 = 0$, $K_2 = \frac{a}{2\omega}$.

Therefore

$$x_p(t) = \frac{at \sin \omega t}{2\omega}$$

The general solution is the complementary function plus the particular integral:

$$x(t) = x_h(t) + x_p(t) = \frac{at\sin\omega t}{2\omega} + c_1\cos\omega t + c_2\sin\omega t$$

Remarks on this case. Now $s = i\omega$ is a simple root of P, so the guess is multiplied by t and matching gives $K_1 = 0$, $K_2 = \frac{a}{2\omega}$:

$$x(t) = c_1\cos\omega t + c_2\sin\omega t + \frac{a}{2\omega}\,t\sin\omega t$$

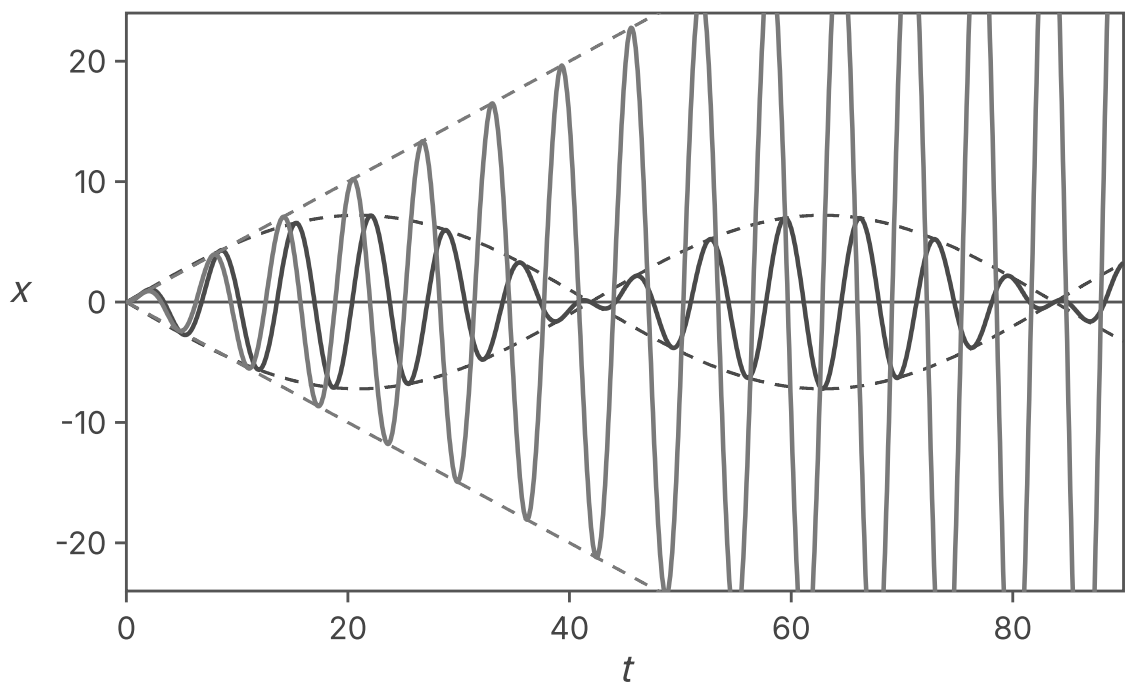
The last term oscillates at the natural frequency with an amplitude $\frac{a}{2\omega}t$ that **grows linearly in time**: the response is unbounded (Robinson Sec. 15.1.2). For zero initial data, $c_1 = c_2 = 0$ (the particular integral and its derivative $\frac{a}{2\omega}(\sin\omega t + \omega t\cos\omega t)$ vanish at $t = 0$), so $x(t) = \frac{a}{2\omega}\,t\sin\omega t$.

Case C: resonance as the limit of beats

Let $\alpha \to \omega$ in the beat solution of Case A, which is $0/0$ at $\alpha = \omega$. L'Hopital's rule (differentiate numerator and denominator with respect to α, at fixed t):

$$\lim_{\alpha\to\omega}\frac{a(\cos\alpha t - \cos\omega t)}{\omega^2 - \alpha^2} = \lim_{\alpha\to\omega}\frac{-at\sin\alpha t}{-2\alpha} = \frac{a\,t\sin\omega t}{2\omega}$$

which is exactly the resonant solution of Case B: the linear growth is the limit of beats whose period and peak both tend to infinity.



Response of $\ddot{x} + x = \cos \alpha t$ from rest ($\omega = 1$): beats for $\alpha = 0.85$ (blue, bounded, envelope dashed) and resonance for $\alpha = 1$ (pink, amplitude $t/2$).

PHASE 4: INTERPRETATION AND CHECKS

Frequency ratio	**Particular integral**	**Behaviour of the general solution**
$\alpha \neq \omega$	$\dfrac{a}{\omega^2 - \alpha^2} \cos \alpha t$	bounded superposition of two sinusoids; beats if the data are those of rest; large amplitude near $\alpha \approx \omega$
$\alpha = \omega$	$\dfrac{a}{2\omega} t \sin \omega t$	amplitude grows linearly: resonance
$\alpha \to 0$	$\dfrac{a}{\omega^2}$	static deflection a/ω^2 under a constant force (Hooke's law)
$\alpha \to \infty$	$\approx -\dfrac{a}{\alpha^2} \cos \alpha t$	response tends to 0: inertia dominates

- There is no damping in this model, so the transient x_h never dies: unlike Exercise 4, the steady state is not isolated from the initial data.
- Dimensions: $[a] = \mathrm{m/s^2}$, $[\omega^2 - \alpha^2] = \mathrm{s^{-2}}$, so $[a/(\omega^2 - \alpha^2)] = \mathrm{m}$ and for the resonant term $[a\,t/(2\omega)] = (\mathrm{m/s^2})(\mathrm{s})/(\mathrm{s^{-1}}) = \mathrm{m}$ (radians are dimensionless).
- Engineering reading: a structure with natural frequency ω must not be forced at (or near) ω for long; real systems have damping, which keeps the amplitude finite (Exercise 4).

EXERCISE 4

Damped Forced Oscillator: Amplitude and Phase of the Steady State

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Frequency Response • Amplitude M • Phase φ

STATEMENT

Consider the oscillating system of the previous exercise, but now in the presence of a non-negligible viscous drag so that Newton's second law reads

$$\ddot{x} + \lambda \dot{x} + \omega^2 x = a \cos \alpha t,$$

in the underdamped case in which $\lambda^2 < 4\omega^2$. Show that $x_p(t) = M \cos(\alpha t - \phi)$ is a particular solution, finding its amplitude M and phase ϕ in terms of the system parameters. Discuss how the relative ratio between the system response and the external forcing, M/a, behaves with the external frequency α and compare with the zero-drag ($\lambda = 0$) case.

PHASE 1: SETUP AND HYPOTHESES

Parameters: $\lambda > 0$ (damping rate, s^{-1}), $\omega > 0$ (natural frequency), $\alpha > 0$ (forcing frequency), $a > 0$ (forcing amplitude per unit mass). Underdamped: $\lambda^2 < 4\omega^2$, so the characteristic roots are $-\frac{\lambda}{2} \pm i\sigma$ with $\sigma = \frac{1}{2}\sqrt{4\omega^2 - \lambda^2} > 0$ (Topic 3, Sec. 4.3) and the complementary function is $x_h = e^{-\lambda t/2}(c_1 \cos \sigma t + c_2 \sin \sigma t)$, which decays. Only x_p survives at long times.

PHASE 2: FORMULATION: WHAT HAS TO BE SHOWN

We must find constants M and ϕ such that $x_p = M \cos(\alpha t - \phi)$ satisfies $L[x_p] = a \cos \alpha t$. Because $\cos(\alpha t - \phi)$ is a shifted cosine, the cosine and sine terms of the guess of Topic 3, Sec. 6.1 are packed in one amplitude and one phase. The subtraction formulas $\cos(\alpha t - \phi) = \cos \alpha t \cos \phi + \sin \alpha t \sin \phi$ and $\sin(\alpha t - \phi) = \sin \alpha t \cos \phi - \cos \alpha t \sin \phi$ will be used.

PHASE 3: DERIVATION

Substitution

Differentiate with the chain rule ($\frac{d}{dt}(\alpha t - \phi) = \alpha$):

$$\dot{x}_p = -M\alpha \sin(\alpha t - \phi), \qquad \ddot{x}_p = -M\alpha^2 \cos(\alpha t - \phi)$$

Therefore

$$L[x_p] = \ddot{x}_p + \lambda \dot{x}_p + \omega^2 x_p = M\left[(\omega^2 - \alpha^2)\cos(\alpha t - \phi) - \lambda\alpha \sin(\alpha t - \phi)\right]$$

Expand the shifted functions and collect cosines and sines of αt:

$$L[x_p] = M\left[(\omega^2 - \alpha^2)\cos\phi + \lambda\alpha \sin\phi\right]\cos\alpha t + M\left[(\omega^2 - \alpha^2)\sin\phi - \lambda\alpha\cos\phi\right]\sin\alpha t$$

This must equal $a\cos\alpha t + 0 \cdot \sin\alpha t$ for every t; $\cos\alpha t$ and $\sin\alpha t$ are linearly independent, so both coefficients must match:

$$M\left[(\omega^2 - \alpha^2)\cos\phi + \lambda\alpha\sin\phi\right] = a \quad (1)$$
$$M\left[(\omega^2 - \alpha^2)\sin\phi - \lambda\alpha\cos\phi\right] = 0 \quad (2)$$

Phase

$M \neq 0$, because $M = 0$ would turn equation (1) into $0 = a > 0$. Hence equation (2) gives $(\omega^2 - \alpha^2)\sin\phi = \lambda\alpha\cos\phi$, i.e.

$$\tan\phi = \frac{\lambda\alpha}{\omega^2 - \alpha^2}$$

To fix the quadrant, introduce $R = \sqrt{(\omega^2 - \alpha^2)^2 + \lambda^2\alpha^2} > 0$ and the angle ϕ defined by

$$\cos\phi = \frac{\omega^2 - \alpha^2}{R}, \qquad \sin\phi = \frac{\lambda\alpha}{R}$$

(these satisfy $\cos^2\phi + \sin^2\phi = \frac{(\omega^2-\alpha^2)^2 + \lambda^2\alpha^2}{R^2} = 1$ and equation (2): $(\omega^2 - \alpha^2)\frac{\lambda\alpha}{R} - \lambda\alpha\frac{\omega^2-\alpha^2}{R} = 0$).
Because $\sin\phi = \lambda\alpha/R > 0$, $\phi \in (0, \pi)$: $\phi = \operatorname{arccot}\frac{\omega^2-\alpha^2}{\lambda\alpha}$, i.e. $\phi = \arctan\frac{\lambda\alpha}{\omega^2-\alpha^2}$ for $\alpha < \omega$ and $\phi = \pi + \arctan\frac{\lambda\alpha}{\omega^2-\alpha^2}$ for $\alpha > \omega$.

Amplitude

Substitute $\cos\phi$ and $\sin\phi$ in equation (1):

$$M\left[(\omega^2-\alpha^2)\frac{\omega^2-\alpha^2}{R}+\lambda\alpha\frac{\lambda\alpha}{R}\right]=M\,\frac{(\omega^2-\alpha^2)^2+\lambda^2\alpha^2}{R}=M\,R=a$$

$$\boxed{M=\frac{a}{\sqrt{(\omega^2-\alpha^2)^2+\lambda^2\alpha^2}},\qquad \phi=\operatorname{atan2}\big(\lambda\alpha,\ \omega^2-\alpha^2\big)}$$

so $x_p=M\cos(\alpha t-\phi)$ is indeed a particular solution. It agrees with the sine-cosine form of Topic 3, Eq. 15.5: $M\cos\phi=\frac{a(\omega^2-\alpha^2)}{R^2}$ and $M\sin\phi=\frac{a\lambda\alpha}{R^2}$ are exactly the coefficients D and C found there.

Ratio of response to forcing

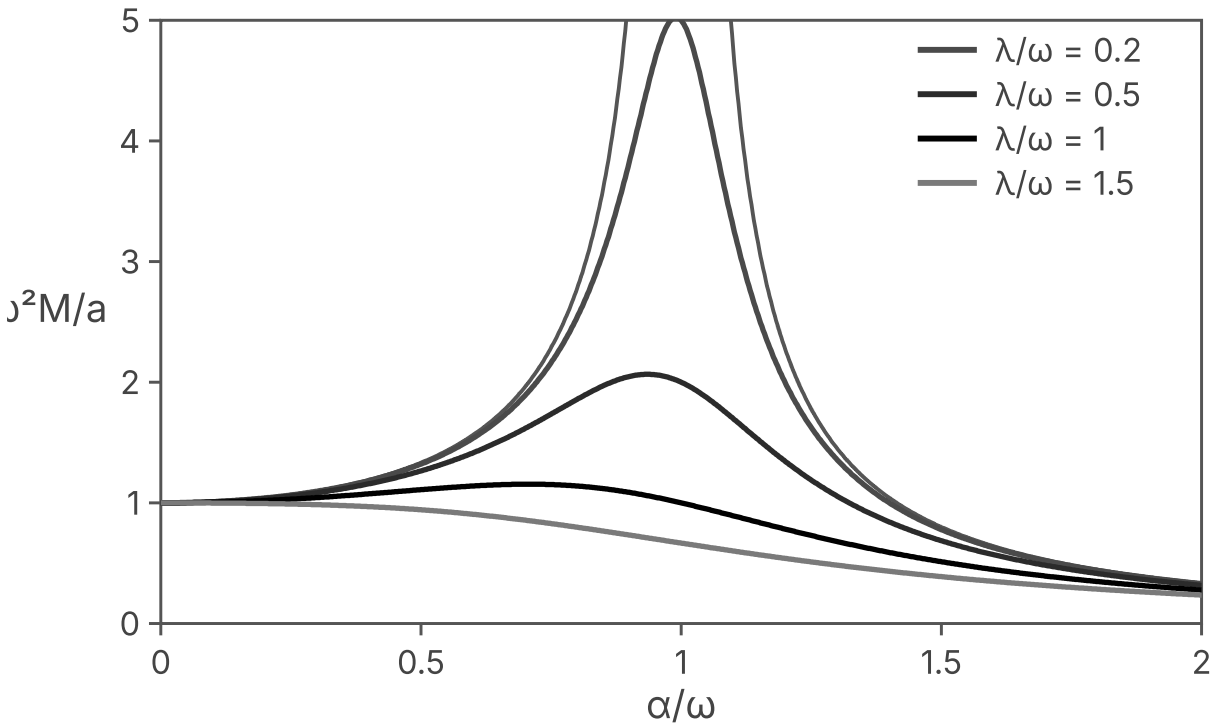
$$\frac{M}{a}=\frac{1}{\sqrt{(\omega^2-\alpha^2)^2+\lambda^2\alpha^2}}=:F(\alpha)$$

Differentiate the radicand $D(\alpha)=(\omega^2-\alpha^2)^2+\lambda^2\alpha^2$ with the chain rule: $D'(\alpha)=-4\alpha(\omega^2-\alpha^2)+2\lambda^2\alpha=2\alpha\left(2\alpha^2-2\omega^2+\lambda^2\right)$. The critical points are $\alpha=0$ and $\alpha_r^2=\omega^2-\frac{\lambda^2}{2}$, which is real only if $\lambda^2<2\omega^2$.

- **Case $\lambda^2<2\omega^2$ (light damping).** D has a minimum at $\alpha_r=\sqrt{\omega^2-\lambda^2/2}<\omega$ and F a maximum. There $\omega^2-\alpha_r^2=\frac{\lambda^2}{2}$, so $D(\alpha_r)=\frac{\lambda^4}{4}+\lambda^2\left(\omega^2-\frac{\lambda^2}{2}\right)=\lambda^2\left(\omega^2-\frac{\lambda^2}{4}\right)$ and $\left(\frac{M}{a}\right)_{\max}=\frac{1}{\lambda\sqrt{\omega^2-\lambda^2/4}}=\frac{1}{\lambda\sigma}$: finite, large for small λ, and attained slightly below ω (pseudo-resonance).
- **Case $2\omega^2\le\lambda^2<4\omega^2$.** $D'(\alpha)\ge0$ for $\alpha>0$: no peak, and M/a decreases monotonically from its value at $\alpha=0$.
- **Limits.** $\alpha\to0$: $M/a\to1/\omega^2$ (static deflection under a constant force); $\alpha=\omega$: $M/a=1/(\lambda\omega)$; $\alpha\to\infty$: $M/a\approx1/\alpha^2\to0$.
- **Phase.** ϕ increases continuously from 0 at $\alpha=0$ through $\pi/2$ at $\alpha=\omega$ (there $\cos\phi=0$) to π as $\alpha\to\infty$: the response lags the forcing by a quarter period exactly at the natural frequency.

Comparison with the undamped case

For $\lambda=0$, $M/a=\frac{1}{|\omega^2-\alpha^2|}$, which tends to $+\infty$ as $\alpha\to\omega$ (resonance, Exercise 3), and ϕ jumps discontinuously from 0 (for $\alpha<\omega$) to π (for $\alpha>\omega$): the response is in phase below ω and in antiphase above. Any $\lambda>0$ regularises this: the peak becomes finite, $1/(\lambda\sigma)$, and the phase jump is smoothed over a band of width $\sim\lambda$ around ω. As $\lambda\to0$, $\alpha_r\to\omega$ and the peak height $1/(\lambda\sigma)\to\infty$, recovering resonance.



Amplitude ratio $\omega^2 M/a$ against α/ω for $\lambda/\omega = 0.2, 0.5, 1, 1.5$ (blue to pink) and the undamped limit (dashed, $\lambda = 0$).

PHASE 4: INTERPRETATION AND CHECKS

RESULT

$x_p(t) = M\cos(\alpha t - \phi)$ with $M = \dfrac{a}{\sqrt{(\omega^2 - \alpha^2)^2 + \lambda^2\alpha^2}}$ and $\phi = \operatorname{atan2}(\lambda\alpha, \omega^2 - \alpha^2) \in (0, \pi)$.

The general solution is $x = e^{-\lambda t/2}(c_1\cos\sigma t + c_2\sin\sigma t) + M\cos(\alpha t - \phi)$: the transient decays and the steady state at the forcing frequency remains, independent of the initial data.

- Numerical check of (1) and (2) at $\omega = 2, \lambda = 1, \alpha = 1, a = 1$: $R = \sqrt{9+1} = \sqrt{10}$, $M = 1/\sqrt{10}$, $\cos\phi = 3/\sqrt{10}$, $\sin\phi = 1/\sqrt{10}$, and $M[3\cos\phi + 1\cdot\sin\phi] = \frac{1}{\sqrt{10}}\cdot\frac{9+1}{\sqrt{10}} = 1 = a$ (checked symbolically in `scripts/verify_topic3_topic4.py`).
- Dimensions: $[\omega^2 - \alpha^2] = [\lambda\alpha] = \mathrm{s}^{-2}$, so $[R] = \mathrm{s}^{-2}$ and $[M] = [a]/[R] = \mathrm{m}/\mathrm{s}^2\cdot\mathrm{s}^2 = \mathrm{m}$; ϕ is dimensionless.
- Condition $\lambda^2 < 4\omega^2$ is only needed for the form of x_h; the formulas for M and ϕ hold for every $\lambda > 0$.

EXERCISE 5

Higher-Order Inhomogeneous Equations with Constant Coefficients

Original statement · probls_ch3-ii_2627.pdf, p. 2

Third and Fourth Order • Repeated Roots • Undetermined Coefficients

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STATEMENT

Working by analogy with the second order case, find the general solution of the following inhomogeneous higher-order linear equations with constant coefficients:

- (i) $\dfrac{d^3x}{dt^3} - 6\dfrac{d^2x}{dt^2} + 11\dfrac{dx}{dt} - 6x = e^{-t}$,
- (ii) $y''' - 3y' + 2 = \sin x$,
- (iii) $\dfrac{d^4x}{dt^4} - 4\dfrac{d^3x}{dt^3} + 8\dfrac{d^2x}{dt^2} - 8\dfrac{dx}{dt} + 4x = \sin t$.

PHASE 1: SETUP, CLASSIFICATION AND A REMARK ON THE STATEMENT

All three are linear, constant-coefficient, inhomogeneous equations of order 3, 3 and 4. By the higher-order theory (Robinson Sec. 16.1, Topic 3 Sec. 3 box), the general solution of $a_n x^{(n)} + \cdots + a_0 x = f$ is $x = x_h + x_p$, where x_h contains n arbitrary constants (the solution space of the homogeneous equation has dimension n) and the method of undetermined coefficients gives x_p exactly as for $n = 2$. Hypotheses: coefficients constant, f continuous.

REMARK ON THE STATEMENT OF (II)

The sheet prints $y''' - 3y' + 2 = \sin x$. Read literally this is the linear equation $y''' - 3y' = \sin x - 2$ (the constant 2 moves to the right-hand side), which is perfectly solvable. The other two items and the form $L[y] = \ldots$ suggest that the term $2y$ may have been intended, but that is only a guess from the pattern of the sheet. We therefore solve **both readings in full**: $y''' - 3y' + 2y = \sin x$ (the guessed one) and the literal $y''' - 3y' = \sin x - 2$, each with its own general solution, so that the answer is complete whichever the author meant.

PHASE 2: METHOD

Characteristic polynomial $P(k) = a_n k^n + \cdots + a_0$. A root k of multiplicity m gives $t^j e^{kt}$ for $j = 0, \ldots, m-1$; a complex pair $\rho \pm i\omega$ of multiplicity m gives $t^j e^{\rho t}\cos \omega t$ and $t^j e^{\rho t}\sin \omega t$ for $j = 0, \ldots, m-1$ (Topic 3, Sec. 3 box, Robinson Sec. 16.1). The particular integral uses the guess table of Topic 3, Sec. 5, with the factor t^{m_s} when the exponent s of the forcing is a root of P of multiplicity m_s. The derivatives needed are now up to order n.

PHASE 3: DERIVATION

(i) $x''' - 6x'' + 11x' - 6x = e^{-t}$

The homogeneous equation \dddot$x - 6\ddot{x} + 11\dot{x} - 6x = 0$ has constant coefficients, so we try $x = e^{kt}$ (Robinson Sec. 12.1 and 16.1): by the chain rule each derivative multiplies by k, and dividing by $e^{kt} \neq 0$ leaves the characteristic equation

$$P(k) = k^3 - 6k^2 + 11k - 6 = 0$$

Factorising, $P(k) = (k-3)\,(k-2)\,(k-1)$, so the roots are

$$k = 1, \qquad k = 2, \qquad k = 3$$

A real root k of multiplicity m gives $t^j e^{kt}$ for $j = 0, \ldots, m-1$; a complex pair $\rho \pm i\omega$ gives $t^j e^{\rho t}\cos \omega t$ and $t^j e^{\rho t}\sin \omega t$ (Euler's formula, Robinson Sec. 12.3). The fundamental set is e^{t}, e^{2t}, e^{3t}, so the complementary function is

$$x_h(t) = c_1 e^{t} + c_2 e^{2t} + c_3 e^{3t}$$

The forcing e^{-t} is an exponential with exponent $s = -1$; we test it in the characteristic polynomial: $P(-1) = -24$.

Since $P(s) \neq 0$, no term of the standard guess solves the homogeneous equation, so the standard guess is used as it stands.

Guess for the particular integral, with undetermined coefficients:

$$x_p(t) = K_1 e^{-t}$$

Differentiate term by term (product rule on $t^j e^{\sigma t}$, chain rule on $\cos \omega t$ and $\sin \omega t$):

$$x_p'(t) = -K_1 e^{-t}$$
$$x_p''(t) = K_1 e^{-t}$$
$$x_p^{(3)}(t) = -K_1 e^{-t}$$

Substitute into $L[x_p] = x_p^{(3)} - 6x_p'' + 11x_p' - 6x_p$ and collect the terms that multiply the same function:

$$L[x_p] = -24K_1\, e^{-t}$$

This must equal the forcing $f(t) = e^{-t}$ for every t. The functions on the left are linearly independent, so the coefficient of each must match the coefficient of the same function in f:

$$e^{-t}: \quad -24K_1 = 1$$

The linear system is

$$-24K_1 = 1 \quad (1)$$

$-24K_1 = 1$ contains one unknown only, so $K_1 = -\frac{1}{24}$.

Hence $K_1 = -\frac{1}{24}$.

Therefore

$$x_p(t) = -\frac{e^{-t}}{24}$$

The general solution is the complementary function plus the particular integral:

$$x(t) = x_h(t) + x_p(t) = c_1 e^{t} + c_2 e^{2t} + c_3 e^{3t} - \frac{e^{-t}}{24}$$

(ii) $y''' - 3y' + 2y = \sin x$ **(independent variable** x**)**

The homogeneous equation $y^{(3)} - 3y' + 2y = 0$ has constant coefficients, so we try $y = e^{kx}$ (Robinson Sec. 12.1 and 16.1): by the chain rule each derivative multiplies by k, and dividing by $e^{kx} \neq 0$ leaves the characteristic equation

$$P(k) = k^3 - 3k + 2 = 0$$

Factorising, $P(k) = (k-1)^2\,(k+2)$, so the roots are

$$k = -2, \qquad k = 1 \text{ (multiplicity 2)}$$

A real root k of multiplicity m gives $x^j e^{kx}$ for $j = 0, \dots, m-1$; a complex pair $\rho \pm i\omega$ gives $x^j e^{\rho x}\cos\omega x$ and $x^j e^{\rho x}\sin\omega x$ (Euler's formula, Robinson Sec. 12.3). The fundamental set is e^{-2x}, e^{x}, xe^{x}, so the complementary function is

$$y_h(x) = c_1 e^{-2x} + c_2 e^{x} + c_3 x e^{x}$$

The forcing $\sin x$ is a sinusoidal term: it is the real or imaginary part of $e^{(i)x}$, so we test $s = i$ in the characteristic polynomial: $P(i) = 2 - 4i$.

Since $P(s) \neq 0$, no term of the standard guess solves the homogeneous equation, so the standard guess is used as it stands.

Guess for the particular integral, with undetermined coefficients:

$$y_p(x) = K_1 \cos x + K_2 \sin x$$

Differentiate term by term (product rule on $x^j e^{\sigma x}$, chain rule on $\cos\omega x$ and $\sin\omega x$):

$$\begin{aligned} y_p'(x) &= -K_1 \sin x + K_2 \cos x \\ y_p''(x) &= -K_1 \cos x - K_2 \sin x \\ y_p^{(3)}(x) &= K_1 \sin x - K_2 \cos x \end{aligned}$$

Substitute into $L[y_p] = y_p^{(3)} - 3y_p' + 2y_p$ and collect the terms that multiply the same function:

$$L[y_p] = (2K_1 - 4K_2)\,\cos x + (4K_1 + 2K_2)\,\sin x$$

This must equal the forcing $f(x) = \sin x$ for every x. The functions on the left are linearly independent, so the coefficient of each must match the coefficient of the same function in f:

$$\begin{aligned}\cos x: &\quad 2K_1 - 4K_2 = 0\\ \sin x: &\quad 4K_1 + 2K_2 = 1\end{aligned}$$

The linear system is

$$\begin{aligned}2K_1 - 4K_2 &= 0 \quad (1)\\ 4K_1 + 2K_2 &= 1 \quad (2)\end{aligned}$$

Solve $2K_1 - 4K_2 = 0$ for K_1: $K_1 = 2K_2$.

Substitute in $4K_1 + 2K_2 = 1$: $10K_2 = 1$.

$10K_2 = 1$ contains one unknown only, so $K_2 = \frac{1}{10}$.

Back-substitution: $K_1 = 2K_2 = \frac{1}{5}$.

Hence $K_1 = \frac{1}{5}, K_2 = \frac{1}{10}$.

Therefore

$$y_p(x) = \frac{\sin x}{10} + \frac{\cos x}{5}$$

The general solution is the complementary function plus the particular integral:

$$y(x) = y_h(x) + y_p(x) = c_1 e^{-2x} + c_2 e^{x} + c_3 x e^{x} + \frac{\sin x}{10} + \frac{\cos x}{5}$$

Literal reading of the sheet: $y''' - 3y' = \sin x - 2$

The homogeneous equation $y^{(3)} - 3y' = 0$ has constant coefficients, so we try $y = e^{kx}$ (Robinson Sec. 12.1 and 16.1): by the chain rule each derivative multiplies by k, and dividing by $e^{kx} \neq 0$ leaves the characteristic equation

$$P(k) = k^3 - 3k = 0$$

Factorising, $P(k) = k\left(k^2 - 3\right)$, so the roots are

$$k = -\sqrt{3}, \qquad k = 0, \qquad k = \sqrt{3}$$

A real root k of multiplicity m gives $x^j e^{kx}$ for $j = 0, \dots, m-1$; a complex pair $\rho \pm i\omega$ gives $x^j e^{\rho x}\cos\omega x$ and $x^j e^{\rho x}\sin\omega x$ (Euler's formula, Robinson Sec. 12.3). The fundamental set is $e^{-\sqrt{3}x}$, 1, $e^{\sqrt{3}x}$, so the complementary function is

$$y_h(x) = c_1 e^{-\sqrt{3}x} + c_2 + c_3 e^{\sqrt{3}x}$$

The forcing is a sum of terms. By linearity (Robinson Eq. 14.7) a particular integral is the sum of particular integrals of the separate terms, each found in turn.

Particular integral for the term $\sin x$

The forcing $\sin x$ is a sinusoidal term: it is the real or imaginary part of $e^{(i)x}$, so we test $s = i$ in the characteristic polynomial: $P(i) = -4i$.

Since $P(s) \neq 0$, no term of the standard guess solves the homogeneous equation, so the standard guess is used as it stands.

Guess for the particular integral, with undetermined coefficients:

$$y_p(x) = K_1 \cos x + K_2 \sin x$$

Differentiate term by term (product rule on $x^j e^{\sigma x}$, chain rule on $\cos\omega x$ and $\sin\omega x$):

$$\begin{aligned} y_p'(x) &= -K_1 \sin x + K_2 \cos x \\ y_p''(x) &= -K_1 \cos x - K_2 \sin x \\ y_p^{(3)}(x) &= K_1 \sin x - K_2 \cos x \end{aligned}$$

Substitute into $L[y_p] = y_p^{(3)} - 3y_p'$ and collect the terms that multiply the same function:

$$L[y_p] = -4K_2\ \cos x + 4K_1\ \sin x$$

This must equal the forcing $f(x) = \sin x$ for every x. The functions on the left are linearly independent, so the coefficient of each must match the coefficient of the same function in f:

$$\begin{aligned} \cos x: &\quad -4K_2 = 0 \\ \sin x: &\quad 4K_1 = 1 \end{aligned}$$

The linear system is

$$-4K_2 = 0 \quad (1)$$
$$4K_1 = 1 \quad (2)$$

$-4K_2 = 0$ contains one unknown only, so $K_2 = 0$.

$4K_1 = 1$ contains one unknown only, so $K_1 = \frac{1}{4}$.

Hence $K_1 = \frac{1}{4}, K_2 = 0$.

Therefore

$$y_p(x) = \frac{\cos x}{4}$$

Particular integral for the term -2

The forcing -2 is an exponential with exponent $s = 0$; we test it in the characteristic polynomial: $P(0) = 0$.

Since $P(s) = 0$ and s is a root of multiplicity 1, the standard guess would contain terms of the complementary function. By the rule of thumb (Robinson Sec. 14.5, p. 139) the guess is multiplied by x^1.

Guess for the particular integral, with undetermined coefficients:

$$y_p(x) = K_1 x$$

Differentiate term by term (product rule on $x^j e^{\sigma x}$, chain rule on $\cos \omega x$ and $\sin \omega x$):

$$y_p'(x) = K_1$$
$$y_p''(x) = 0$$
$$y_p^{(3)}(x) = 0$$

Substitute into $L[y_p] = y_p^{(3)} - 3y_p'$ and collect the terms that multiply the same function:

$$L[y_p] = -3K_1$$

This must equal the forcing $f(x) = -2$ for every x. The functions on the left are linearly independent, so the coefficient of each must match the coefficient of the same function in f:

$$1: \quad -3K_1 = -2$$

The linear system is

$$-3K_1 = -2 \quad (1)$$

$-3K_1 = -2$ contains one unknown only, so $K_1 = \frac{2}{3}$.

Hence $K_1 = \frac{2}{3}$.

Therefore

$$y_p(x) = \frac{2x}{3}$$

Adding the particular integrals,

$$y_p(x) = \frac{2x}{3} + \frac{\cos x}{4}$$

The general solution is the complementary function plus the particular integral:

$$y(x) = y_h(x) + y_p(x) = c_1 e^{-\sqrt{3}x} + c_2 + c_3 e^{\sqrt{3}x} + \frac{2x}{3} + \frac{\cos x}{4}$$

(iii) $x^{(4)} - 4x''' + 8x'' - 8x' + 4x = \sin t$

The homogeneous equation $x^{(4)} - 4\dddot{x} + 8\ddot{x} - 8\dot{x} + 4x = 0$ has constant coefficients, so we try $x = e^{kt}$ (Robinson Sec. 12.1 and 16.1): by the chain rule each derivative multiplies by k, and dividing by $e^{kt} \neq 0$ leaves the characteristic equation

$$P(k) = k^4 - 4k^3 + 8k^2 - 8k + 4 = 0$$

Factorising, $P(k) = (k-1-i)^2\,(k-1+i)^2$, so the roots are

$$k = 1 - i \text{ (multiplicity 2)}, \qquad k = 1 + i \text{ (multiplicity 2)}$$

A real root k of multiplicity m gives $t^j e^{kt}$ for $j = 0, \dots, m-1$; a complex pair $\rho \pm i\omega$ gives $t^j e^{\rho t}\cos\omega t$ and $t^j e^{\rho t}\sin\omega t$ (Euler's formula, Robinson Sec. 12.3). The fundamental set is $e^t\cos t$, $e^t\sin t$, $te^t\cos t$, $te^t\sin t$, so the complementary function is

$$x_h(t) = c_1 e^t\cos t + c_2 e^t\sin t + c_3 te^t\cos t + c_4 te^t\sin t$$

The forcing $\sin t$ is a sinusoidal term: it is the real or imaginary part of $e^{(i)t}$, so we test $s = i$ in the characteristic polynomial: $P(i) = -3 - 4i$.

Since $P(s) \neq 0$, no term of the standard guess solves the homogeneous equation, so the standard guess is used as it stands.

Guess for the particular integral, with undetermined coefficients:

$$x_p(t) = K_1\cos t + K_2\sin t$$

Differentiate term by term (product rule on $t^j e^{\sigma t}$, chain rule on $\cos\omega t$ and $\sin\omega t$):

$$\begin{aligned} x_p'(t) &= -K_1\sin t + K_2\cos t \\ x_p''(t) &= -K_1\cos t - K_2\sin t \\ x_p^{(3)}(t) &= K_1\sin t - K_2\cos t \\ x_p^{(4)}(t) &= K_1\cos t + K_2\sin t \end{aligned}$$

Substitute into $L[x_p] = x_p^{(4)} - 4x_p^{(3)} + 8x_p'' - 8x_p' + 4x_p$ and collect the terms that multiply the same function:

$$L[x_p] = (-3K_1 - 4K_2)\,\cos t + (4K_1 - 3K_2)\,\sin t$$

This must equal the forcing $f(t) = \sin t$ for every t. The functions on the left are linearly independent, so the coefficient of each must match the coefficient of the same function in f:

$$\begin{aligned} \cos t: &\quad -3K_1 - 4K_2 = 0 \\ \sin t: &\quad 4K_1 - 3K_2 = 1 \end{aligned}$$

The linear system is

$$-3K_1 - 4K_2 = 0 \quad (1)$$
$$4K_1 - 3K_2 = 1 \quad (2)$$

Solve $-3K_1 - 4K_2 = 0$ for K_1: $K_1 = -\frac{4K_2}{3}$.

Substitute in $4K_1 - 3K_2 = 1$: $-\frac{25K_2}{3} = 1$.

$-\frac{25K_2}{3} = 1$ contains one unknown only, so $K_2 = -\frac{3}{25}$.

Back-substitution: $K_1 = -\frac{4K_2}{3} = \frac{4}{25}$.

Hence $K_1 = \frac{4}{25}$, $K_2 = -\frac{3}{25}$.

Therefore

$$x_p(t) = -\frac{3\sin t}{25} + \frac{4\cos t}{25}$$

The general solution is the complementary function plus the particular integral:

$$x(t) = x_h(t) + x_p(t) = c_1 e^t \cos t + c_2 e^t \sin t + c_3 t e^t \cos t + c_4 t e^t \sin t - \frac{3\sin t}{25} + \frac{4\cos t}{25}$$

PHASE 4: INTERPRETATION AND CHECKS

Item	General solution
(i)	$c_1 e^t + c_2 e^{2t} + c_3 e^{3t} - \frac{e^{-t}}{24}$
(ii), intended	$y(x) = c_1 e^{-2x} + c_2 e^x + c_3 x e^x + \frac{\sin x}{10} + \frac{\cos x}{5}$
(ii), literal reading	$y(x) = c_1 e^{-\sqrt{3}x} + c_2 + c_3 e^{\sqrt{3}x} + \frac{2x}{3} + \frac{\cos x}{4}$
(iii)	$c_1 e^t \cos t + c_2 e^t \sin t + c_3 t e^t \cos t + c_4 t e^t \sin t - \frac{3\sin t}{25} + \frac{4\cos t}{25}$

- The number of arbitrary constants equals the order (3, 3, 4), as the dimension of the solution space requires (Robinson Sec. 16.2).
- (i): three distinct real roots $1, 2, 3$, none equal to the exponent $s = -1$ of the forcing, hence no factor t.
- (ii): $P(k) = k^3 - 3k + 2 = (k-1)^2(k+2)$ has the **double** root 1, giving e^x and xe^x; the forcing exponent is $s = \pm i$, not a root.
- (iii): $P(k) = k^4 - 4k^3 + 8k^2 - 8k + 4 = (k^2 - 2k + 2)^2 = ((k-1)^2 + 1)^2$ has the **double** complex pair $1 \pm i$, giving $e^t \cos t$, $e^t \sin t$, $te^t \cos t$, $te^t \sin t$; the forcing exponent is $\pm i$, not a root.

- Each particular integral was substituted back into its equation symbolically (script `scripts/verify_topic3_topic4.py`); the residual vanishes identically.

EXERCISE 6

Variation of Constants: Five Particular Solutions

Original statement · probls_ch3-ii_2627.pdf, p. 2

Variation of Parameters • Euler Equation • Integral Answers

STATEMENT

Use the method of variation of constants to find a particular solution for the following equations:

- (i) $y'' - y' - 6y = e^x$;
- (ii) $\ddot{x} - x = t^{-1}$ (leave the answer as an integral);
- (iii) $y'' + 4y = \cot 2x$. Use that $\int \csc x\, dx = \ln|\csc x - \cot x|$;
- (iv) $t^2\ddot{x} - 2x = t^3$ (to find the solutions of the homogeneous equation, try solutions of the form $x = t^k$ with k a parameter);
- (v) $\ddot{x} - 4\dot{x} = \tan t$ (leave your answer as an integral).

PHASE 1: SETUP AND HYPOTHESES

Each equation is linear, second-order, inhomogeneous. Variation of constants (Lagrange; Robinson Ch. 18, Topic 3 Sec. 8) needs: (a) the equation in **standard form** $\ddot{x} + p\dot{x} + qx = g$ with the coefficient of the highest derivative equal to 1; (b) two linearly independent solutions x_1, x_2 of the homogeneous equation; (c) g continuous on an interval I where the coefficients are continuous. Then, on I,

$$x_p = -x_1 \int \frac{x_2\, g}{W}\, dt + x_2 \int \frac{x_1\, g}{W}\, dt, \qquad W = x_1\dot{x}_2 - x_2\dot{x}_1 \neq 0 \quad \text{(Robinson Eq. 18.8)}$$

Item	Standard form	Interval I where g and the coefficients are continuous
(i)	$y'' - y' - 6y = e^x$	$\mathbb{R}$
(ii)	$\ddot{x} - x = 1/t$	$t > 0$ (or $t < 0$)
(iii)	$y'' + 4y = \cot 2x$	$0 < x < \pi/2$ (between zeros of $\sin 2x$)
(iv)	$\ddot{x} - \frac{2}{t^2}x = t$ (divide by t^2)	$t > 0$ (or $t < 0$)

Item	Standard form	Interval I where g and the coefficients are continuous
(v)	$\ddot{x} - 4\dot{x} = \tan t$	$-\pi/2 < t < \pi/2$

PHASE 2: FORMULATION: THE DERIVATION OF THE FORMULA, RECALLED

Replace the constants of $x_h = c_1 x_1 + c_2 x_2$ by functions: $x = u_1 x_1 + u_2 x_2$. Choose $\dot{u}_1 x_1 + \dot{u}_2 x_2 = 0$ (18.7), so that $\dot{x} = u_1 \dot{x}_1 + u_2 \dot{x}_2$; substituting in the equation and using $L[x_1] = L[x_2] = 0$ leaves $\dot{u}_1 \dot{x}_1 + \dot{u}_2 \dot{x}_2 = g$. Cramer's rule on the system with determinant W gives

$$\dot{u}_1 = -\frac{x_2\, g}{W}, \qquad \dot{u}_2 = \frac{x_1\, g}{W}, \qquad x_p = u_1 x_1 + u_2 x_2$$

Constants of integration only add multiples of x_1, x_2, so they may be dropped.

PHASE 3: DERIVATION

(i) $y'' - y' - 6y = e^x$

Homogeneous equation: $y = e^{kx}$ gives $k^2 - k - 6 = (k-3)(k+2) = 0$, so $k = 3, -2$.

The homogeneous solutions are $y_1 = e^{3x}$ and $y_2 = e^{-2x}$. Their Wronskian is

$$W = y_1\, y_2' - y_2\, y_1' = (e^{3x})(-2e^{-2x}) - (e^{-2x})(3e^{3x}) = -5e^{x}$$

With $g = e^x$ (equation in standard form, coefficient of y'' equal to 1), Cramer's rule gives $u_1' = -\dfrac{y_2\, g}{W}$ and $u_2' = \dfrac{y_1\, g}{W}$:

$$u_1' = -\frac{(e^{-2x})(e^{x})}{-5e^{x}} = \frac{e^{-2x}}{5}, \qquad u_2' = \frac{(e^{3x})(e^{x})}{-5e^{x}} = -\frac{e^{3x}}{5}$$

Integrate (antiderivatives of exponentials, $\int e^{kx}dx = e^{kx}/k$):

$$u_1 = \int \frac{e^{-2x}}{5}\,dx = -\frac{e^{-2x}}{10}, \qquad u_2 = \int -\frac{e^{3x}}{5}\,dx = -\frac{e^{3x}}{15}$$

$$y_p = u_1 y_1 + u_2 y_2 = \left(-\frac{e^{-2x}}{10}\right)e^{3x} + \left(-\frac{e^{3x}}{15}\right)e^{-2x} = -\frac{e^{x}}{6} = -\frac{e^{x}}{6}$$

Check: $y_p' = -\frac{e^x}{6}$, $y_p'' = -\frac{e^x}{6}$, so $y_p'' - y_p' - 6y_p = (-\frac{e^x}{6}) - (-\frac{e^x}{6}) - 6(-\frac{e^x}{6}) = e^x$.

(ii) $\ddot{x} - x = t^{-1}$ (answer as an integral)

Homogeneous equation: $k^2 - 1 = 0$, so $k = \pm 1$: $x_1 = e^t$, $x_2 = e^{-t}$.

The homogeneous solutions are $x_1 = e^t$ and $x_2 = e^{-t}$. Their Wronskian is

$$W = x_1\,x_2' - x_2\,x_1' = (e^t)(-e^{-t}) - (e^{-t})(e^t) = -2$$

With $g = \frac{1}{t}$ (equation in standard form, coefficient of x'' equal to 1), Cramer's rule gives $u_1' = -\dfrac{x_2\,g}{W}$ and $u_2' = \dfrac{x_1\,g}{W}$:

$$u_1' = -\frac{(e^{-t})(\frac{1}{t})}{-2} = \frac{e^{-t}}{2t}, \qquad u_2' = \frac{(e^t)(\frac{1}{t})}{-2} = -\frac{e^t}{2t}$$

The antiderivatives of e^{-s}/s and e^s/s are not elementary (they are exponential integrals). We keep them as integrals from a fixed base point $t_0 > 0$ (the interval where $g = 1/t$ is continuous, $t > 0$), which is a legitimate antiderivative by the fundamental theorem of calculus:

$$u_1(t) = \frac{1}{2}\int_{t_0}^{t} \frac{e^{-s}}{s}\,ds, \qquad u_2(t) = -\frac{1}{2}\int_{t_0}^{t} \frac{e^{s}}{s}\,ds$$

$$\boxed{x_p(t) = \frac{e^t}{2}\int_{t_0}^{t} \frac{e^{-s}}{s}\,ds - \frac{e^{-t}}{2}\int_{t_0}^{t} \frac{e^{s}}{s}\,ds \quad (t, t_0 > 0)}$$

Combining the exponentials inside one integral, $x_p(t) = \dfrac{1}{2}\displaystyle\int_{t_0}^{t} \frac{e^{t-s} - e^{-(t-s)}}{s}\,ds = \int_{t_0}^{t} \frac{\sinh(t-s)}{s}\,ds$.

Verification by differentiation (Leibniz rule)

Let $I_1(t) = \int_{t_0}^{t} e^{-s}s^{-1}ds$ and $I_2(t) = \int_{t_0}^{t} e^{s}s^{-1}ds$, so $I_1' = e^{-t}/t$ and $I_2' = e^t/t$. Then $x_p = \frac{1}{2}(e^tI_1 - e^{-t}I_2)$ and, by the product rule,

$$\dot{x}_p = \frac{1}{2}\left(e^tI_1 + e^{-t}I_2\right) + \frac{1}{2}\left(e^t\frac{e^{-t}}{t} - e^{-t}\frac{e^t}{t}\right) = \frac{1}{2}\left(e^tI_1 + e^{-t}I_2\right)$$

$$\ddot{x}_p = \frac{1}{2}\left(e^t I_1 - e^{-t} I_2\right) + \frac{1}{2}\left(e^t \frac{e^{-t}}{t} + e^{-t}\frac{e^t}{t}\right) = x_p + \frac{1}{t}$$

so $\ddot{x}_p - x_p = 1/t$, as required.

(iii) $y'' + 4y = \cot 2x$

Homogeneous equation: $k^2 + 4 = 0$, $k = \pm 2i$, so $y_1 = \cos 2x$, $y_2 = \sin 2x$.

Wronskian (chain rule: $(\cos 2x)' = -2\sin 2x$, $(\sin 2x)' = 2\cos 2x$):

$$W = \cos 2x \cdot 2\cos 2x - \sin 2x \cdot (-2\sin 2x) = 2(\cos^2 2x + \sin^2 2x) = 2$$

Cramer's rule with $g = \cot 2x = \cos 2x / \sin 2x$:

$$u_1' = -\frac{\sin 2x \, \cot 2x}{2} = -\frac{\cos 2x}{2}, \qquad u_2' = \frac{\cos 2x \, \cot 2x}{2} = \frac{\cos^2 2x}{2\sin 2x}$$

For u_2' use $\cos^2 2x = 1 - \sin^2 2x$: $u_2' = \dfrac{1 - \sin^2 2x}{2\sin 2x} = \dfrac{1}{2}(\csc 2x - \sin 2x)$. Integrate with the substitution $s = 2x$, $ds = 2\,dx$, and the hint $\int \csc s \, ds = \ln|\csc s - \cot s|$:

$$u_1 = -\frac{1}{2}\int \cos 2x \, dx = -\frac{1}{2}\cdot\frac{\sin 2x}{2} = -\frac{\sin 2x}{4}$$

$$\int \csc 2x \, dx = \frac{1}{2}\int \csc s \, ds = \frac{1}{2}\ln|\csc 2x - \cot 2x|, \qquad \int \sin 2x \, dx = -\frac{\cos 2x}{2}$$

$$u_2 = \frac{1}{2}\left[\frac{1}{2}\ln|\csc 2x - \cot 2x| + \frac{\cos 2x}{2}\right] = \frac{1}{4}\ln|\csc 2x - \cot 2x| + \frac{\cos 2x}{4}$$

Assemble $y_p = u_1 y_1 + u_2 y_2$; the terms in $\sin 2x \cos 2x$ cancel:

$$y_p = -\frac{\sin 2x \cos 2x}{4} + \left[\frac{1}{4}\ln|\csc 2x - \cot 2x| + \frac{\cos 2x}{4}\right]\sin 2x = \frac{1}{4}\,\sin 2x \,\ln|\csc 2x - \cot 2x|$$

Since $\csc 2x - \cot 2x = \dfrac{1-\cos 2x}{\sin 2x} = \dfrac{2\sin^2 x}{2\sin x\cos x} = \tan x$, an equivalent form is $y_p = \frac{1}{4}\sin 2x\ \ln|\tan x|$.

Check by differentiating $y_p = \frac{1}{4}\sin 2x\,\ell$ with $\ell = \ln\tan x$, $\ell' = \frac{\sec^2 x}{\tan x} = \frac{2}{\sin 2x}$: $y_p' = \frac{1}{2}\cos 2x\,\ell + \frac{1}{4}\sin 2x\cdot\frac{2}{\sin 2x} = \frac{1}{2}\cos 2x\,\ell + \frac{1}{2}$; $y_p'' = -\sin 2x\,\ell + \frac{1}{2}\cos 2x\cdot\frac{2}{\sin 2x} = -\sin 2x\,\ell + \cot 2x$; hence $y_p'' + 4y_p = -\sin 2x\,\ell + \cot 2x + \sin 2x\,\ell = \cot 2x$.

(iv) $t^2\ddot{x} - 2x = t^3$

Homogeneous equation $t^2\ddot{x} - 2x = 0$. Try $x = t^k$ (chain/power rule): $\dot{x} = kt^{k-1}$, $\ddot{x} = k(k-1)t^{k-2}$, so $t^2\ddot{x} - 2x = [k(k-1)-2]t^k = (k^2-k-2)t^k = (k-2)(k+1)t^k = 0$, hence $k = 2, -1$: $x_1 = t^2$, $x_2 = t^{-1}$ $(t > 0)$.

Standard form: divide by t^2 (coefficient of $\ddot{x}$ must be 1): $\ddot{x} - \dfrac{2}{t^2}x = \dfrac{t^3}{t^2} = t$, so $g(t) = t$ (**not** t^3).

The homogeneous solutions are $x_1 = t^2$ and $x_2 = \frac{1}{t}$. Their Wronskian is

$$W = x_1\,x_2' - x_2\,x_1' = (t^2)(-\frac{1}{t^2}) - (\frac{1}{t})(2t) = -3$$

With $g = t$ (equation in standard form, coefficient of x'' equal to 1), Cramer's rule gives $u_1' = -\dfrac{x_2\,g}{W}$ and $u_2' = \dfrac{x_1\,g}{W}$:

$$u_1' = -\frac{(\frac{1}{t})(t)}{-3} = \frac{1}{3}, \qquad u_2' = \frac{(t^2)(t)}{-3} = -\frac{t^3}{3}$$

$$u_1 = \int \frac{1}{3}\,dt = \frac{t}{3}, \qquad u_2 = \int -\frac{t^3}{3}\,dt = -\frac{t^4}{12}$$

$$x_p = u_1x_1 + u_2x_2 = \left(\frac{t}{3}\right)t^2 + \left(-\frac{t^4}{12}\right)t^{-1} = \frac{t^3}{4} = \frac{t^3}{4}$$

Check in the original equation: $\ddot{x}_p = \frac{3t}{2}$, so $t^2\ddot{x}_p - 2x_p = \frac{3t^3}{2} - \frac{t^3}{2} = t^3$.

(v) $\ddot{x} - 4\dot{x} = \tan t$ **(answer as an integral)**

Homogeneous equation: $k^2 - 4k = k(k-4) = 0$, $k = 0, 4$: $x_1 = 1$, $x_2 = e^{4t}$.

The homogeneous solutions are $x_1 = 1$ and $x_2 = e^{4t}$. Their Wronskian is

$$W = x_1\,x_2' - x_2\,x_1' = (1)(4e^{4t}) - (e^{4t})(0) = 4e^{4t}$$

With $g = \tan t$ (equation in standard form, coefficient of x'' equal to 1), Cramer's rule gives $u_1' = -\dfrac{x_2\,g}{W}$ and $u_2' = \dfrac{x_1\,g}{W}$:

$$u_1' = -\frac{(e^{4t})(\tan t)}{4e^{4t}} = -\frac{\tan t}{4}, \qquad u_2' = \frac{(1)(\tan t)}{4e^{4t}} = \frac{e^{-4t}\tan t}{4}$$

The first integral is elementary: $\int -\frac{1}{4}\tan t\,dt = \frac{1}{4}\ln|\cos t|$ (since $\frac{d}{dt}\ln\cos t = -\tan t$). The second, $\int e^{-4s}\tan s\,ds$, is not elementary, so it is left as an integral from a base point $t_0 \in (-\pi/2, \pi/2)$:

$$u_1(t) = \frac{1}{4}\ln\cos t, \qquad u_2(t) = \frac{1}{4}\int_{t_0}^{t} e^{-4s}\tan s\,ds$$

$$\boxed{x_p(t) = \frac{1}{4}\ln\cos t + \frac{e^{4t}}{4}\int_{t_0}^{t} e^{-4s}\tan s\,ds \qquad (|t|, |t_0| < \pi/2)}$$

Verification by differentiation

With $I(t) = \int_{t_0}^{t} e^{-4s}\tan s\,ds$, $I' = e^{-4t}\tan t$. Product rule: $\dot{x}_p = -\frac{1}{4}\tan t + e^{4t}I + \frac{e^{4t}}{4}e^{-4t}\tan t = e^{4t}I$ (the two tangent terms cancel). Again, $\ddot{x}_p = 4e^{4t}I + e^{4t}e^{-4t}\tan t = 4\dot{x}_p + \tan t$, i.e. $\ddot{x}_p - 4\dot{x}_p = \tan t$.

PHASE 4: INTERPRETATION AND CHECKS

Item	Particular solution x_p	Domain
(i)	$-\dfrac{e^{x}}{6}$	$\mathbb{R}$
(ii)	$\dfrac{e^{t}}{2}\displaystyle\int_{t_0}^{t}\frac{e^{-s}}{s}ds - \frac{e^{-t}}{2}\int_{t_0}^{t}\frac{e^{s}}{s}ds$	$t > 0$
(iii)	$\dfrac{1}{4}\sin 2x\,\ln\lvert\csc 2x - \cot 2x\rvert = \dfrac{1}{4}\sin 2x\,\ln\lvert\tan x\rvert$	$0 < x < \pi/2$
(iv)	$\dfrac{t^3}{4}$	$t > 0$
(v)	$\dfrac{1}{4}\ln\cos t + \dfrac{e^{4t}}{4}\displaystyle\int_{t_0}^{t} e^{-4s}\tan s\,ds$	$\lvert t\rvert < \pi/2$

- The general solution of each equation is $x_h + x_p$ with the two homogeneous solutions used above.

- (i) and (iv) could also have been found by undetermined coefficients (Ke^x and Kt^3); both methods agree ($-\frac{1}{6}e^x$ and $\frac{1}{4}t^3$), a useful cross-check of the Wronskian bookkeeping. (iii) and (v) have forcings ($\cot$, $\tan$) outside the family of undetermined coefficients, which is the reason for the method.
- (iv) is an Euler equation: the substitution $x = t^k$ turns it into an algebraic equation for k, and its standard form needs the division by t^2 before applying the formula.
- Wronskians: $-5e^x$ in (i) (consistent with Abel's identity, $W = W_0 e^x$ since $p = -1$), -2 in (ii), 2 in (iii), -3 in (iv) and $4e^{4t}$ in (v) (Abel: $p = -4$, $W = W_0 e^{4t}$); none vanishes on its interval.
- The sheet works in dimensionless variables, so no unit check is attached to the forcings (t^3, e^x, $\cot 2x$, $\tan t$ are pure numbers); the check that matters is that each particular solution reproduces its forcing exactly, which was done symbolically.
- Every particular solution (and the integral forms, via the differentiation above) was verified symbolically in `scripts/verify_topic3_topic4.py`.

EXERCISE 7

Splitting the Solution of an IVP into Homogeneous and Inhomogeneous Parts

Original statement · probls_ch3-ii_2627.pdf, p. 2

Initial Value Problem • Superposition • Uniqueness

STATEMENT

Show that the solution of the initial value problem

$$L[y] = y'' + p(t)y' + q(t)y = g(t), \qquad y(t_0) = y_0, \quad y'(t_0) = y_0'$$

can be written as $y(t) = u(t) + v(t)$, where u and v are solutions of two different initial value problems:

$$\begin{aligned} L[u] &= 0, \qquad u(t_0) = y_0, \quad u'(t_0) = y_0', \\ L[v] &= g(t), \quad v(t_0) = 0, \quad v'(t_0) = 0. \end{aligned}$$

PHASE 1: SETUP AND HYPOTHESES

Hypotheses: p, q, g are continuous on an open interval $I \ni t_0$. Under these hypotheses the existence and uniqueness theorem (Robinson Thm 11.1, Topic 3 Sec. 1; for linear equations the solution exists on the whole of I) states that every IVP $L[w] = h, w(t_0) = w_0, w'(t_0) = w_1$ has exactly one solution on I. Define $L[y] = y'' + p\,y' + q\,y$; L is a **linear operator**: $L[\alpha y_1 + \beta y_2] = \alpha L[y_1] + \beta L[y_2]$ (Robinson Eq. 11.11).

PHASE 2: FORMULATION

Let y be the unique solution of the original IVP, u the unique solution of the homogeneous IVP with the data of y, and v the unique solution of the inhomogeneous IVP with zero data (both exist on I by the theorem). The claim is $y = u + v$ on I. Proof strategy: show that $u + v$ solves the **original** IVP; uniqueness then forces $u + v = y$.

PHASE 3: DERIVATION

Step 1: $u+v$ solves the differential equation

$$L[u+v] = L[u] + L[v] = 0 + g(t) = g(t) \quad \text{on } I,$$

by linearity of L (differentiation is linear: $(u+v)' = u' + v'$ and $(u+v)'' = u'' + v''$, so $L[u+v] = u'' + v'' + p(u'+v') + q(u+v) = (u'' + pu' + qu) + (v'' + pv' + qv)$).

Step 2: $u+v$ satisfies the initial conditions

$$(u+v)(t_0) = u(t_0) + v(t_0) = y_0 + 0 = y_0, \qquad (u+v)'(t_0) = u'(t_0) + v'(t_0) = y_0' + 0 = y_0'$$

Step 3: uniqueness

Both y and $u+v$ solve the IVP $L[w] = g, w(t_0) = y_0, w'(t_0) = y_0'$ on I. By uniqueness (Robinson Thm 11.1), $y = u + v$ on I. An equivalent argument that avoids quoting the theorem twice: $w = y - (u+v)$ satisfies $L[w] = g - g = 0$ with $w(t_0) = 0$ and $w'(t_0) = 0$; the zero function solves the same homogeneous IVP, so $w \equiv 0$ by uniqueness. ■

Where the hypotheses were used

- Linearity of L: Steps 1 and the alternative argument.
- Uniqueness: Step 3. Existence of u and v on all of I: needed so that $u+v$ is defined where y is; it holds because L is linear with continuous coefficients.

PHASE 4: INTERPRETATION AND CHECKS

- Meaning: u is the **free response** (the system left to itself with the given initial state) and v the **forced response** from rest (zero state). This is the decomposition used in control and vibration analysis, e.g. u = transient from initial displacement and velocity, v = response to the external force from a state of rest.
- It reduces any IVP with general data to two simpler ones: the homogeneous one with the data, and the inhomogeneous one with zero data. Variation of constants gives v directly: with x_1, x_2 independent homogeneous solutions and W their Wronskian, $v(t) = \int_{t_0}^{t} \dfrac{x_1(s)x_2(t) - x_1(t)x_2(s)}{W(s)}\, g(s)\, ds$ (check: $v(t_0) = 0$ and $v'(t_0) = 0$ because the integrand vanishes at $s = t$ and the extra boundary term is 0).
- Example (Exercise 1(ii) of this sheet, $\ddot{x} - 4\dot{x} = t^2, x(0) = 2, \dot{x}(0) = 0$): $u = 2$ (constant; it solves $\ddot{u} - 4\dot{u} = 0$ with $u(0) = 2, \dot{u}(0) = 0$) and $v = x_p + c_1 + c_2 e^{4t}$ with $v(0) = \dot{v}(0) = 0$; their sum is the solution found there.

EXERCISE 8

Reduction of Order for the Inhomogeneous Equation

Original statement · probls_ch3-ii_2627.pdf, p. 2

Reduction of Order • Integrating Factor • General Solution

STATEMENT

The so-called method of reduction of order can be used to find one solution of the inhomogeneous equation $L[y] = y'' + p(t)y' + q(t)y = g(t)$, provided one solution $y_1(t)$ of the corresponding homogeneous equation is known. Indeed, writing $y = v(t)y_1(t)$, show that $y(t)$ satisfies $L[y] = g$ if v is a solution of

$$y_1(t)v'' + [2y_1' + p(t)y_1]\,v' = g(t).$$

This is a first order linear equation for $v'(t)$. Solving this equation, integrating the result, and then multiplying by $y_1(t)$ leads to the general solution of the problem for $y(t)$.

PHASE 1: SETUP AND HYPOTHESES

Hypotheses: p, q, g continuous on an interval I; y_1 is a known solution of $L[y_1] = 0$ with $y_1 \neq 0$ on I (so that we can divide by y_1). Unknown: $v(t)$, with $y = vy_1$.

PHASE 2: FORMULATION

The idea is the same as in Topic 3, Sec. 7: promote the constant in Ay_1 to a function, $y = vy_1$. The product rule is used twice:

$$y' = v'y_1 + vy_1', \qquad y'' = v''y_1 + 2v'y_1' + vy_1''$$

PHASE 3: DERIVATION

Step 1: substitute in $L[y]$

$$L[y] = y'' + py' + qy = (v''y_1 + 2v'y_1' + vy_1'') + p\,(v'y_1 + vy_1') + qvy_1$$

Group the terms by v'', v' and v:

$$L[y] = y_1 v'' + (2y_1' + py_1)\,v' + v\underbrace{(y_1'' + py_1' + qy_1)}_{L[y_1]=0}$$

The bracket multiplying v is $L[y_1]$, which is zero because y_1 solves the homogeneous equation; this is why the undifferentiated v disappears. Therefore

$$L[y] = y_1 v'' + (2y_1' + py_1)\,v'$$

and $L[y] = g$ holds if and only if $y_1 v'' + [2y_1' + py_1]v' = g$, which is the equation of the statement. ■

Step 2: solve the first-order equation for $w = v'$
Divide by $y_1 \neq 0$ and put $w = v'$:

$$w' + \left(\frac{2y_1'}{y_1} + p\right) w = \frac{g}{y_1}$$

This is a linear first-order equation (Topic 2). The integrating factor is $\mu = \exp\int\left(\frac{2y_1'}{y_1} + p\right)dt = y_1^2\,e^{P(t)}$, with $P(t) = \int p\,dt$ (because $\int \frac{2y_1'}{y_1}dt = 2\ln|y_1|$). The product rule gives $\frac{d}{dt}(\mu w) = \mu w' + \mu' w = \mu\left(w' + \left(\frac{2y_1'}{y_1} + p\right)w\right) = \mu\frac{g}{y_1}$:

$$\frac{d}{dt}\left(y_1^2 e^P w\right) = y_1 e^P g \quad\Longrightarrow\quad w(t) = \frac{1}{y_1^2 e^P}\left[\int y_1 e^P g\,dt + c_1\right]$$

Step 3: integrate for v and multiply by y_1

$$v(t) = \int w\,dt + c_2, \qquad y(t) = y_1(t)\,v(t) = y_1(t)\left\{c_2 + \int \frac{1}{y_1^2 e^P}\left[c_1 + \int y_1 e^P g\,dt\right] dt\right\}$$

This expression contains two arbitrary constants c_1, c_2. Setting $g = 0$ it reduces to the homogeneous solutions $y = c_2 y_1 + c_1 y_1 \int \frac{dt}{y_1^2 e^P}$ (the second solution of Robinson Exercise 17.7), and the part that does not contain c_1, c_2 is a particular solution of $L[y] = g$. The two homogeneous solutions y_1 and $y_2 = y_1 \int \frac{dt}{y_1^2 e^P}$ are independent: by the product rule $y_2' = y_1' \int \frac{dt}{y_1^2 e^P} + \frac{1}{y_1 e^P}$, so $W = y_1 y_2' - y_2 y_1' = e^{-P} \neq 0$ (the terms with the integral cancel). Hence complementary function plus particular integral, with two free constants, is the general solution (Robinson Sec. 14.1).

PHASE 4: INTERPRETATION AND CHECKS

- The constant c_2 multiplies y_1 itself (the known solution) and c_1 multiplies $y_1 \int \frac{dt}{y_1^2 e^P}$, a second, independent homogeneous solution. Their Wronskian is $W = e^{-P}$, nowhere zero: consistent with Abel's identity $W = W_0 \exp(-\int p)$.
- Check of the Wronskian: $y_2 = y_1 \int \frac{dt}{y_1^2 e^P}$ gives $y_2' = y_1' \int \frac{dt}{y_1^2 e^P} + \frac{1}{y_1 e^P}$, hence $W = y_1 y_2' - y_2 y_1' = \frac{1}{e^P}$ (the $y_1 y_1' \int \ldots$ terms cancel).
- Compared with variation of constants, reduction of order needs only **one** homogeneous solution, at the price of two nested integrations.
- The method applies on any interval where $y_1 \neq 0$; where y_1 vanishes the division is not allowed and the intervals must be treated separately.

EXERCISE 9

A Second Solution of an Euler Equation by Reduction of Order

Original statement · probls_ch3-ii_2627.pdf, p. 2

Reduction of Order • Euler Equation • Wronskian

STATEMENT

> For $t > 0$ the function $y_1(t) = 1/t$ is a solution of $t^2\ddot{y} - 2t\dot{y} - 4y = 0$. Using the previous exercise, find a second, linearly independent solution.

PHASE 1: SETUP AND HYPOTHESES

Domain $t > 0$ (so $y_1 = 1/t \neq 0$). The equation $t^2\ddot{y} - 2t\dot{y} - 4y = 0$ is homogeneous with variable coefficients; dividing by t^2 gives the standard form $\ddot{y} + p\dot{y} + qy = 0$ with $p(t) = -\dfrac{2}{t}$, $q(t) = -\dfrac{4}{t^2}$, continuous for $t > 0$. First check that y_1 is a solution:

$$y_1 = t^{-1},\ \ \dot{y}_1 = -t^{-2},\ \ \ddot{y}_1 = 2t^{-3} \ \Rightarrow\ t^2(2t^{-3}) - 2t(-t^{-2}) - 4t^{-1} = 2t^{-1} + 2t^{-1} - 4t^{-1} = 0\ \checkmark$$

PHASE 2: FORMULATION

Apply Exercise 8 with $g = 0$: write $y = v\,y_1 = v/t$, so that v must satisfy $y_1\ddot{v} + [2\dot{y}_1 + py_1]\dot{v} = 0$. A second independent solution is $y_2 = v\,y_1$ with v non-constant.

PHASE 3: DERIVATION

The equation for v

Compute the coefficient: $2\dot{y}_1 + py_1 = 2(-t^{-2}) + \left(-\dfrac{2}{t}\right)\left(\dfrac{1}{t}\right) = -2t^{-2} - 2t^{-2} = -4t^{-2}$. The equation of Exercise 8 is

$$\frac{1}{t}\,\ddot{v} - \frac{4}{t^2}\,\dot{v} = 0 \quad\Longleftrightarrow\quad \ddot{v} = \frac{4}{t}\,\dot{v} \quad \text{(multiplying by } t\text{)}$$

Solve for $w = \dot{v}$

$\dot{w} = \dfrac{4}{t}w$ is separable (or use the integrating factor $e^{-4\ln t} = t^{-4}$): $\dfrac{d}{dt}(t^{-4}w) = t^{-4}\dot{w} - 4t^{-5}w = t^{-4}\left(\dot{w} - \dfrac{4}{t}w\right) = 0$, so $t^{-4}w = c$ and

$$w = \dot{v} = c\,t^4$$

Integrate and multiply by y_1

$$v = \int ct^4\,dt = \frac{c}{5}t^5 + b \quad\Longrightarrow\quad y = \frac{v}{t} = \frac{c}{5}t^4 + \frac{b}{t}$$

The term in b is the known solution $y_1 = 1/t$; the new, independent solution is (taking $c = 5, b = 0$)

$$\boxed{y_2(t) = t^4}$$

Verification

Substitute: $y_2 = t^4, \dot{y}_2 = 4t^3, \ddot{y}_2 = 12t^2$, so $t^2(12t^2) - 2t(4t^3) - 4t^4 = (12 - 8 - 4)t^4 = 0$ ✓. Wronskian:

$$W[y_1, y_2] = y_1\dot{y}_2 - y_2\dot{y}_1 = \frac{1}{t}(4t^3) - t^4\left(-\frac{1}{t^2}\right) = 4t^2 + t^2 = 5t^2 \neq 0 \quad (t > 0)$$

so y_1 and y_2 are linearly independent on $t > 0$. Abel's identity gives the same dependence on t: $W = W_0 \exp\left(-\int p\,dt\right) = W_0 \exp\left(\int \frac{2}{t}dt\right) = W_0 t^2$, consistent with $5t^2$ ($W_0 = 5$).

PHASE 4: INTERPRETATION AND CHECKS

RESULT

A second independent solution is $y_2 = t^4$, and the general solution on $t > 0$ is $y = c_1 t^{-1} + c_2 t^4$.

- Euler cross-check: for $t^2\ddot{y} - 2t\dot{y} - 4y = 0$ the trial $y = t^k$ gives $k(k-1) - 2k - 4 = k^2 - 3k - 4 = (k-4)(k+1) = 0$, so $k = 4, -1$: the same two solutions, found by a different method.
- Dimensional/scaling check: under $t \to \lambda t$ the Euler equation is invariant and the solutions are homogeneous power functions of degree -1 and 4.
- The general solution is not defined at $t = 0$ (where y_1 blows up and the equation is singular), consistent with the restriction $t > 0$.